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The secretome of senescent monocytes predicts age-related clinical outcomes in humans

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Cellular senescence increases with age and contributes to age-related declines and pathologies. We identified circulating biomarkers of senescence and related them to clinical traits in humans to facilitate future noninvasive assessment of individual senescence burden, and efficacy testing of novel senotherapeutics. Using a nanoparticle-based proteomic workflow, we profiled the senescence-associated secretory phenotype (SASP) in THP-1 monocytes and examined these proteins in 1,060 plasma samples from the Baltimore Longitudinal Study of Aging. Machine-learning models trained on THP-1 monocyte SASP associated SASP signatures with several age-related phenotypes in a test cohort, including body fat composition, blood lipids, inflammatory markers and mobility-related traits, among others. Notably, a subset of SASP-based predictions, including a high-impact SASP panel, were validated in InCHIANTI, an independent aging cohort. These results demonstrate the clinical relevance of the circulating SASP and identify potential senescence biomarkers that could inform future clinical studies.

Recently, senescence of immune cells, including monocytes, have been implicated as a driver of age-related pathologies. Senescent immune cells increase with age and are thought to contribute to age-related sterile inflammation (inflammaging) and susceptibility to infectious diseases[1]. Additionally, elevated levels of senescence in the immune system drives systemic aging and propagates senescence in solid organs, such as the liver, kidney and lung[2]. There is emerging evidence that senescent monocytes accumulate in vivo in humans. A proinflammatory phenotype of monocytes has been attributed to senescence in older adults[3]. Notably, circulating monocytes express a senescence-like signature, as determined by *CDKN1A* positivity and elevated expression of a senescence gene set (SenMayo)[4], are increased in individuals with severe COVID-19 (ref. 5). Moreover, monocytes are promising sources of biomarkers due to their abundance in blood. Despite their potential involvement in inflammaging and biomarker potential, the role of monocyte senescence-associated proteins in aging and their potential as clinical biomarkers are poorly characterized.

Quantifying circulating biomarkers of senescence holds clinical potential for identifying outcomes tied to senescence, noninvasive determinating individual senescence burden for risk stratification, and tracking the effectiveness of senotherapeutics in clinical trials[6,7]. In recent years, proteomic studies have quantitatively profiled SASP

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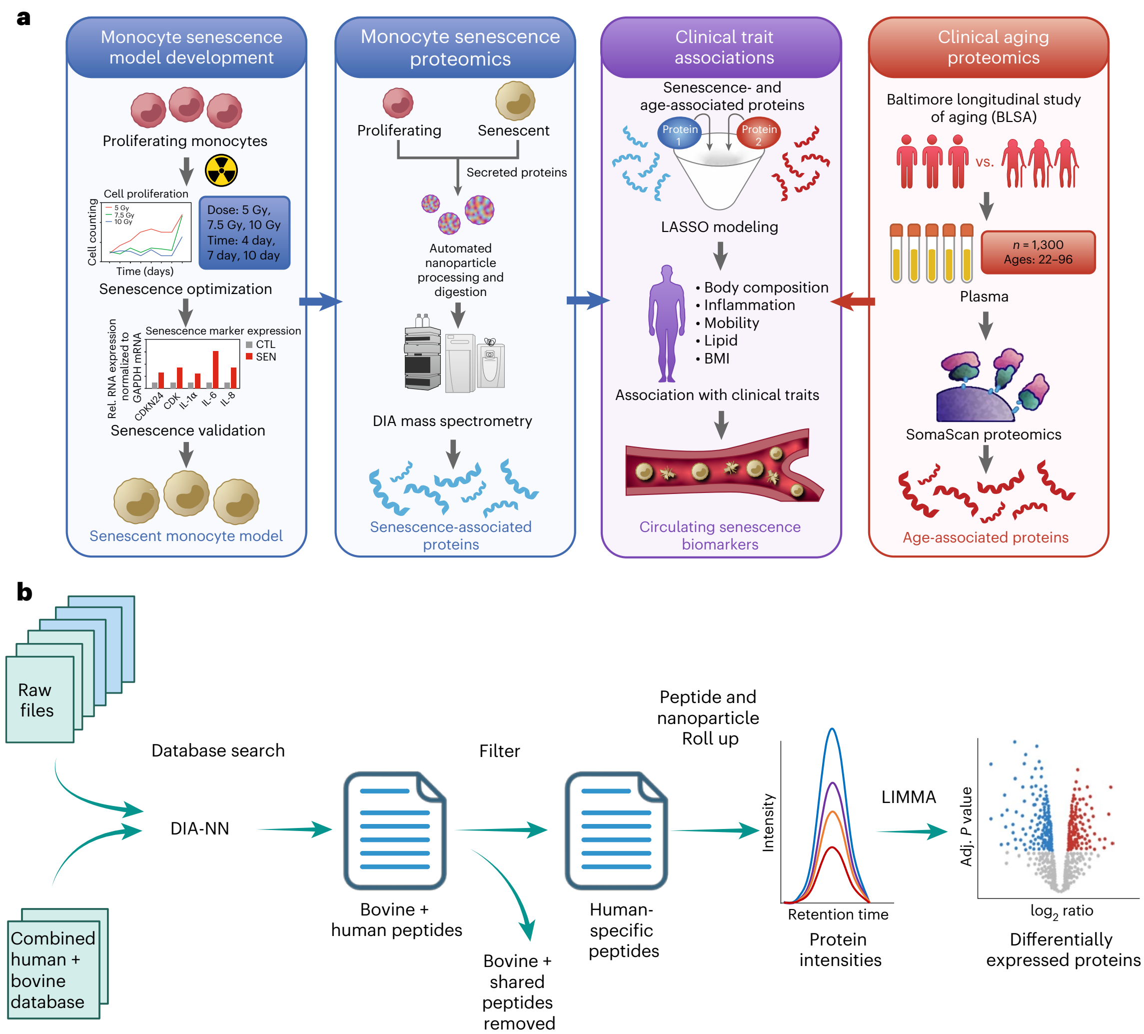

Fig. 1 | Workflow for identification of SASP signatures from the aging plasma proteome. a, An in vitro model of senescent monocytes was developed by exposing THP-1 cells to IR and measuring senescence markers. Further quantitative MS proteomics was performed on the secretome of these cells using the automated nanoparticle processing and digestion platform, Proteograph. Age association of the differentially secreted proteins was evaluated using proteomic and phenotypic data from the BLSA and InCHIANTI aging cohorts. **b**, The analysis pipeline used DIA-NN to identify monocyte secretome followed by filtering out the bovine and shared peptides. Peptide quantities from each nanoparticle were rolled up to proteins to determine the differentially expressed proteins. Image created with BioRender.com.

proteins in circulation and demonstrated their associations with diverse aging-related outcomes in humans[6–8], including mortality, multimorbidity, frailty[9], strength and mobility by leveraging human cohorts, such as InCHIANTI (Invecchiare in Chianti), BLSA (Baltimore Longitudinal Study of Aging), GESTALT (Genetic and Epigenetic Signatures of Translational Aging Laboratory Testing), Lifestyle Interventions for Elders (LIFE) and others[9–15]; however, no biomarker studies have comprehensively characterized the monocyte-specific SASP and evaluated its clinical utility as circulating biomarkers in humans.

The primary goal of this study is to comprehensively profile the monocyte SASP in serum-supplemented culture conditions and evaluate its potential as a circulating biomarker in humans. We adopted an automated nanoparticle-based workflow previously leveraged for analysis of plasma[16] that overcomes the challenges associated with mass spectrometry (MS)-based profiling of serum-supplemented medium, enabling a comprehensive MS analysis of SASP that is not confounded by serum-free culture conditions that are traditionally required for MS-based quantification of SASP from cell-culture experiments[17]. Moreover, we evaluated senescent monocyte signatures in the plasma proteome of BLSA study participants. We identified signatures of SASP, including a high-impact panel, that predict age-related and obesity-associated clinical traits in a test cohort. Remarkably, SASP-based clinical trait associations were replicated in InCHIANTI, an independent aging study.

Results

Optimization of senescence in THP-1 monocytes

To develop a rigorously validated model of cellular senescence in monocytes (Fig. 1a), senescent THP-1 monocytes were generated with multiple protocols and the optimal protocol was selected based on multiple senescence hallmarks, as recommended by the SenNet consortium[7] (Fig. 2), including SASP components, increased lysosomal content and cell-cycle arrest. Senescence was induced in THP-1 monocytes with varying doses of gamma irradiation (IR), 5, 7.5 and 10 Gy, and assessed at various time points after IR. Cell viability measurements at 24-h intervals up to 13 days confirmed all doses were nonlethal (Extended Data Fig. 1a), but cell proliferation was inhibited only after exposure to 7.5 and 10 Gy (Fig. 2a,b and Extended Data Fig. 1b). Further, the expression levels of a panel of senescence mRNAs (*GDF15*, *CDKN2A* (*p16*) and *CDKN1A* (*p21*)) and messenger RNAs encoding SASP factors (such as *IL1A* and *IL6*) were significantly higher in IR-treated cells than proliferating controls (Fig. 2c and Extended Data Fig. 1c–f).

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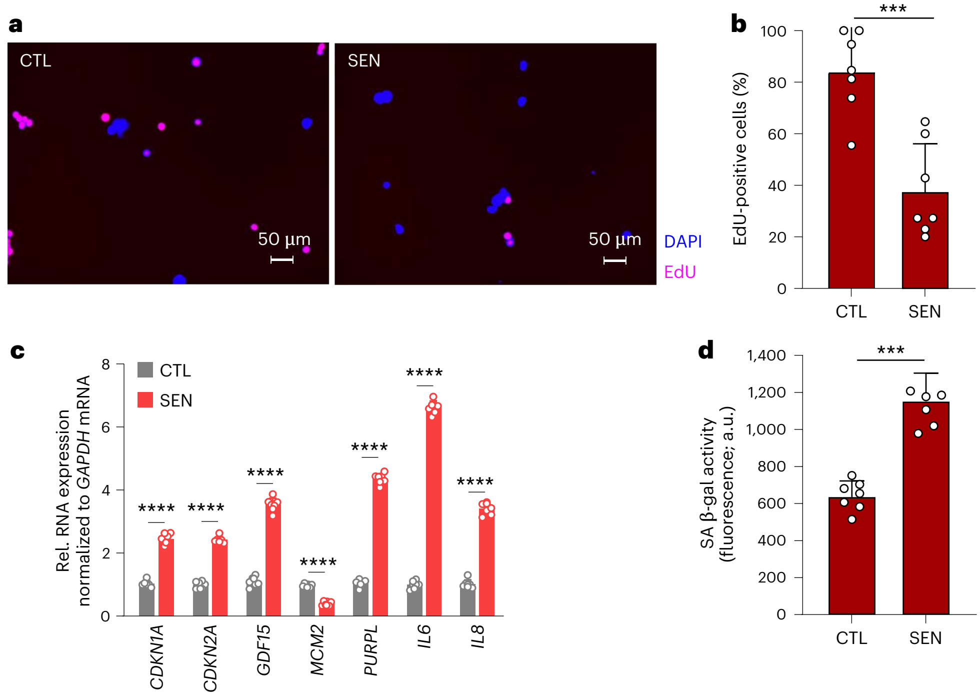

Fig. 2 | Establishing an IR-induced model of senescence in THP-1 monocytes. **a**, Representative fluorescence microscopy images from EdU incorporation assay (blue indicates 4,6-diamidino-2-phenylindole (DAPI)-positive cells and pink indicates 5-ethynyl-2′-deoxyuridine (EdU)-positive cells). **b**, Corresponding quantification bar plots indicating reduced cellular incorporation of EdU by THP-1 cells 7 days after IR exposure ($n = 7$ replicates; $P = 0.00028$). **c**, Bar plots showing increased expression of known senescence markers ($n = 7$ replicates). **d**, Elevated SPiDER-βGal confirming induction of senescence in IR-treated THP-1 cells ($n = 7$ replicates; $P = 3.079 \times 10^{-6}$). Data are represented as mean ± s.d. for n number of replicates in all graphs. Statistical analyses were performed using an unpaired two-tailed Student's t-test (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$).

Elevated levels of SPiDER-βGal in IR-treated cells further confirmed the stable induction of senescence 7 days after exposure to 7.5 Gy (Fig. 2d and Extended Data Fig. 1g). This was also confirmed via conventional senescence-associated β-galactosidase (SA-β-gal) staining (Extended Data Fig. 2a), which revealed higher positivity in IR-treated cells. Further, we confirmed that this protocol did not spuriously induce differentiation of THP-1 monocytes into macrophage-like cells by confirming nonadherence to the culture vessel (Extended Data Fig. 2b) and unchanged expression of cell surface markers CD14 and CD11b, markers of THP-1 differentiation[18] (Extended Data Fig. 2c). Given the stable cell-cycle arrest, viability and expression of senescence-associated mRNAs at this dose, we used 7.5 Gy IR radiation and 7 days in culture in all subsequent experiments.

Identification of the SASP in serum-supplemented culture medium

MS-based proteomic analysis is notoriously challenging in serum-supplemented cell-culture conditions due to the highly dynamic range of serum proteins[19]. In standard cell-culture conditions, fetal bovine serum (FBS) introduces a high concentration of exogenous proteins that hinder the detection of endogenously secreted proteins from cultured cells. To overcome these challenges, we applied an automated platform that utilizes nanoparticle-based enrichment of protein coronas for a deep and unbiased identification of proteins in high-dynamic-range samples such as blood[20,21] (Proteograph XT Assay)[16]. Reasoning that serum-supplemented medium faces essentially the same challenges as serum itself, we applied this instrumentation in combination with a multispecies proteomic analysis (bovine + human) to comprehensively and quantitatively profile the THP-1 monocyte cell SASP (Fig. 1b). Briefly, data-independent acquisition (DIA) MS-based proteomics was conducted on the secretomes of senescent and nonsenescent THP-1 cells processed with the nanoparticle workflow ($n = 14$), as well as matched samples processed with no nanoparticles ($n = 6$) to compare with a standard workflow. Using the standard workflow, 1,492 human proteins (5,686 human peptides) and 3,932 total proteins (24,856 peptides) were identified, and weak separation between the senescent and the nonsenescent secretome was evident by principal-component analysis (PCA) due to interference from the supplemented bovine proteins (Fig. 3a). In contrast, the nanoparticle-based workflow enabled the detection of 5,001 human proteins (14,037 human peptides) and 10,084 total proteins (49,939 peptides) and identified differences in the two secretomes as evident in PCA (Fig. 3b,c); representing a notable improvement over our previous secretome studies[15]. Additionally, although protein expression was positively correlated on the peptide and protein level between neat and nanoparticle-processed samples (Extended Data Fig. 3a,b), the reproducibility of protein measurements was improved after processing (Extended Data Fig. 3c). Additionally, nanoparticle (NP) treatment resulted in an increase in the number (Extended Data Fig. 3d), median intensity (Extended Data Fig. 3e) and the sum intensities (Extended Data Fig. 3f) of human and bovine peptides identified. NP treatment resulted in an increased proportion of identified human peptides compared to bovine peptides, with 5,686 human peptides versus 7,653 bovine peptides (ratio 0.74) with the standard workflow, and 14,037 human peptides versus 11,051 bovine peptides (ratio 1.27) with NP treatment. Any peptides that matched bovine proteins, as well as those shared between human and bovine, were removed, and the remaining human proteins were compared between senescent and nonsenescent secretomes. Senescent secretomes had 3,690 increased proteins and 220 decreased proteins compared to controls (Fig. 3d). Gene Ontology analysis of the top 200 upregulated proteins indicated enrichment of biological processes such as response to interferons, a known senescence-associated pathway, and transmembrane transport (Fig. 3e). Furthermore, a comparison of the THP-1 monocyte cell SASP

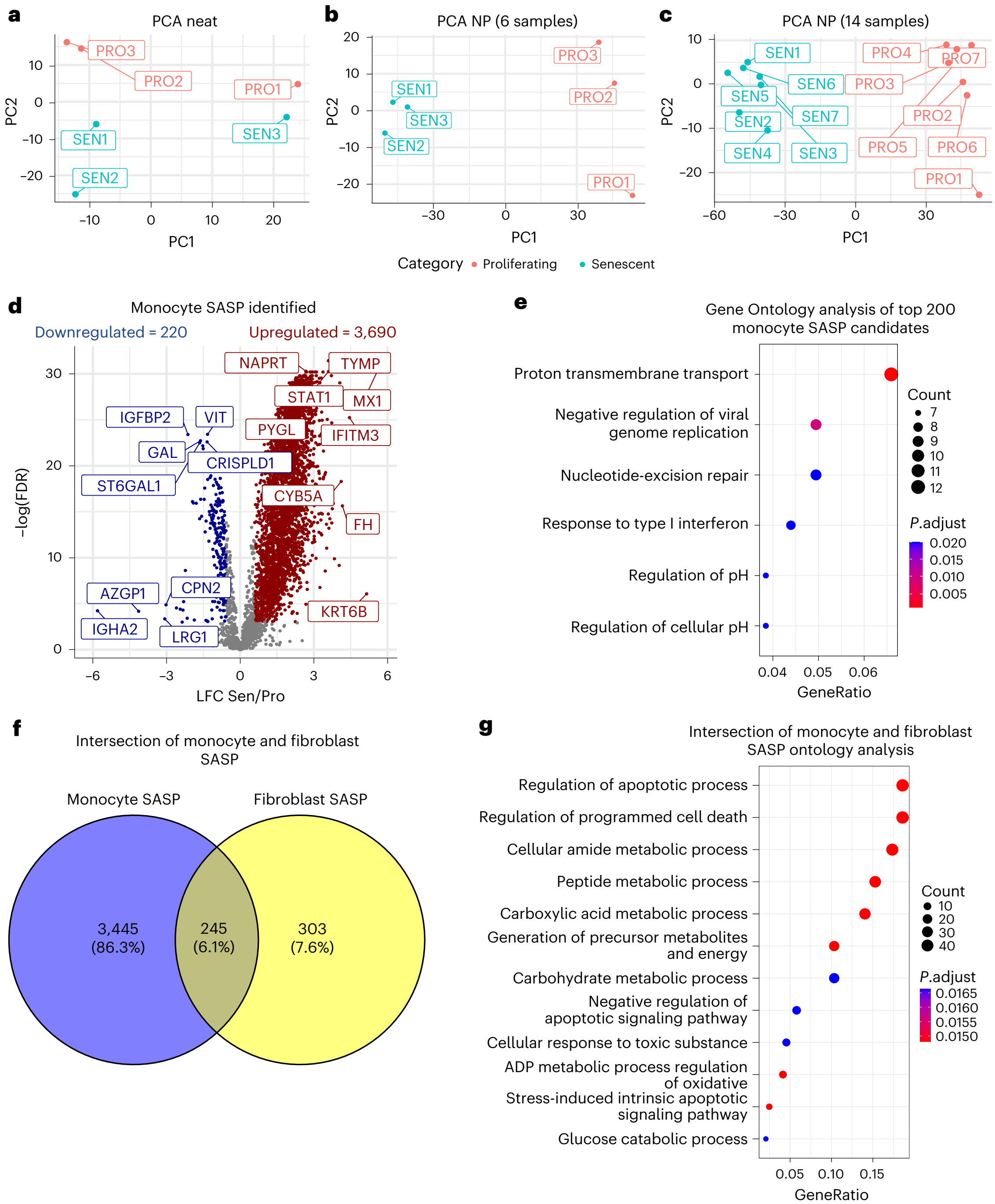

Fig. 3 | Monocyte SASP is Associated with Age in the BLSA. **a**, PCA of Neat samples (protein level). **b**, PCA of 6 NP samples (protein level). **c**, PCA of 14 NP samples (protein levels). For all PCA, only human proteins present in all samples were used (red dots represent proliferating controls while blue represent senescent samples). **d**, THP-1 monocyte SASP were identified using nanoparticle processing on the Proteograph. **e**, The top 200 differentially expressed proteins were used for Ontology analysis for Biological Process ($P < 0.05$, q-value < 0.1 calculated using the Benjamini–Hochberg method. **f**, Overlap between monocyte and fibroblast SASP. **g**, Ontology analysis of overlapping SASP proteins between monocytes and fibroblasts (P adjusted calculated using the Benjamini–Hochberg method).

with that of fibroblasts using the SASP Atlas (http://www.saspatlas.com/)[15] revealed 245 proteins, or 45% of the fibroblast SASP proteome, overlapped when compared to the entire monocyte SASP proteome (Fig. 3f). Ontology analysis reveals that these proteins are involved in multiple biological processes, including cellular detoxification, regulation of apoptotic processes and oxidative stress-induced pathways (Fig. 3g). Notably, these pathways were also reported among the fibroblast core SASP[15]. Full proteomic replicate data, quantification and statistics are available in Supplementary Table 1.

THP-1 monocyte SASP is detected in circulating plasma

The clinical associations of the THP-1 monocyte SASP in circulation were examined in the BLSA. This longitudinal study used multiple biochemical and clinical assessments to examine the physiological and functional changes associated with healthy aging. BLSA participant characteristics are displayed in Table 1. We utilized BLSA plasma proteomic data acquired using the 7K SomaScan Assay (Somalogic), which performs 7,288 protein measurements[22,23]. Out of the 3,413 total monocyte SASP identified by MS, the SomaScan protein panel measured a subset of

Table 1 | Demographic and clinical traits in the BLSA. Data are reported as mean (s.d.)

	All	20 to 40	40 to 60	60 to 70	80+	N
Count	1,060	69	245	576	170	
Age, years	65.7 (14.4)	33.1 (4.6)	51.6 (6)	70.3 (5.5)	83.7 (3)	
Female, %	53.8	53.6	55.9	55.2	45.9	
eGFR	81.4 (17.7)	106.2 (12.4)	94.4 (13.3)	77.1 (14.5)	67.2 (12.8)	1,060
BMI	27 (4.6)	25.2 (4.9)	27.1 (4.8)	27.6 (4.8)	25.7 (3.4)	1,059
Waist size (cm)	87.9 (13.1)	82.2 (12.2)	86.2 (12.6)	89.3 (13.5)	88 (12)	1,046
LDL	106.4 (31.9)	95.6 (28.1)	111.4 (31.2)	106.5 (33)	103.3 (29)	1,052
Triglycerides	99.7 (50.5)	89.4 (43.3)	98.7 (52.8)	102.5 (50.5)	96.1 (49.4)	1,054
CRP	2.6 (4.4)	1.5 (2.1)	2 (3.1)	2.8 (4.3)	3.1 (6.2)	997
IL-6	4.2 (3.5)	3.7 (3.5)	3.6 (2.5)	4.3 (3.5)	4.7 (4.6)	995
Fasting glucose	96.6 (13.5)	90.2 (6.9)	95 (11.5)	98.6 (15.2)	95.1 (11)	979
De_Fat_Trunk	13,298.6 (6,208.6)	10,026.7 (5,926.6)	13,040.1 (6,408.6)	14,333.8 (6,189.1)	11,502.7 (5,143.4)	1,021
De_Fat_Body	26,507.4 (10,343.4)	21,591.9 (10,713.4)	26,390.3 (10,389.2)	28,237.5 (10,484.9)	22,828.3 (7,788.7)	1,021
De_Fat_Arms	2,590.2 (1,079.3)	2,069.5 (1,025.4)	2,583.6 (1,112.5)	2,752.1 (1,097.8)	2,264.5 (826.3)	1,021
De_Fat_Legs	9,726.9 (4,092.8)	8,729.7 (4,340.5)	9,893.1 (3,821.9)	10,222.8 (4,344.1)	8,217.9 (2,916.6)	1,021
CT_Abd_SubFat	27,639.5 (12,750.3)	25,751.6 (18,975.9)	27,062.7 (12,303.6)	29,057 (13,029.9)	23,814.1 (9,508.3)	709
CT_Abd_ViscFat	9,940.6 (5,651.9)	5,511.1 (3,712.2)	7,914.1 (4,619.8)	10,761.3 (5847.2)	10,559.9 (5,447.7)	709
CT_Abd_ViscSub_Prop	0.4 (0.2)	0.2 (0.1)	0.3 (0.2)	0.4 (0.2)	0.5 (0.3)	651
CT_Thigh_SubFat	8,392.1 (4,540.8)	8,094.1 (5,255.4)	8,523.1 (4,329.8)	8,846.6 (4,737.3)	6,832.2 (3,458.5)	899
CT_Thigh_IMFat	1,221.2 (509.6)	895.9 (489.6)	1177.7 (549.3)	1,256.4 (489.2)	1,306.5 (465.6)	897
Total Fat Percent	35.3 (9.4)	29.5 (10.1)	34.3 (9.3)	37 (9.2)	33.6 (8.5)	1,021
BP_Sys	122.5 (18.7)	110.8 (12.5)	118.2 (15.7)	123.8 (18.6)	129.3 (21.8)	883
BP_Dias	66 (9.4)	62.2 (7.1)	68.1 (9.3)	65.9 (9.6)	64.8 (8.6)	883
A1c	5.8 (0.6)	5.3 (0.4)	5.6 (0.6)	5.9 (0.7)	5.8 (0.3)	963
Chair_Stands_Pace_5	0.6 (0.2)	0.7 (0.2)	0.7 (0.2)	0.5 (0.2)	0.5 (0.2)	973
Chair_Stands_Pace_10	0.5 (0.2)	0.7 (0.2)	0.6 (0.2)	0.5 (0.2)	0.5 (0.1)	970
Usual_Gait_Speed (m/s)	1.2 (0.2)	1.3 (0.2)	1.3 (0.2)	1.2 (0.2)	1.1 (0.2)	975
Rapid_Gait_Speed (m/s)	1.8 (0.4)	2.1 (0.3)	2 (0.3)	1.8 (0.4)	1.6 (0.3)	976
Walk_Pace_400_Meter (m/s)	1.6 (0.3)	1.8 (0.2)	1.7 (0.2)	1.6 (0.2)	1.3 (0.2)	927
HDL	61.4 (17.3)	60.1 (15.6)	60.9 (18.1)	61.3 (16.8)	63.1 (18.5)	1,054

Measurements prefixed by De_fat refer to fat content of the respective body part measured by dual X-ray absorptiometry (DEXA) scans in grams. Measurements prefixed by CT refer to fat content of the respective body part measured by computed tomography (CT) scans (mm^2). Abd, abdomen; SubFat, subcutaneous fat; ViscFat, visceral fat; ViscSub_Prop, visceral/subcutaneous fat proportion in abdomen; IMFat, intermuscular fat; BP_Sys, systolic blood pressure; BP_Dias, diastolic blood pressure; A1c, hemoglobin A1c.

1,550 proteins in the BLSA (Fig. 4a and Supplementary Table 2). To evaluate the age-associated clinical relevance of the THP-1 monocyte SASP, we used Spearman correlation to identify only the components of the THP-1 monocyte SASP that were increasing in circulation across the lifespan in the BLSA. Of the THP-1 monocyte SASP in circulating plasma, 308 proteins were upregulated and also positively associated with age (Supplementary Table 2). To evaluate clinical traits associated with SASP regardless of covariates such as age, we utilized the full panel of SASP factors that were measured by the SomaScan 7K assay (1,550 proteins) including covariates such as age, sex and race in downstream clinical trait associations.

SASP signatures predict age-associated clinical outcomes

To identify the most clinically relevant THP-1 monocyte SASPs without model overfitting, least absolute shrinkage and selection operator (LASSO) modeling was performed for unbiased feature selection of proteins that associate with a diverse set of clinical outcomes, while controlling for age, sex, race and kidney function as model covariates. Clinically relevant SASP factors were identified for a panel of clinical traits, including mobility, inflammation, body composition and metabolism-related measurements in the BLSA (Fig. 4b). For each trait, a different subset and number of biologically relevant proteins were identified and subsequently referred to as LASSO-selected proteins (LSPs) (Fig. 4c). The predictive accuracy was assessed in a cohort of 1,060 participants from the BLSA, split into a training dataset (80%) and test dataset (20%). LASSO models, including LSPs and covariates, were significantly correlated with observed values (false discovery rate (FDR) < 0.05) (Fig. 4b) for all clinical traits, with the highest test set trait prediction potential for triglycerides (cor. = 0.8447), high-density lipoprotein (HDL) (cor. = 0.8343), waist size (cor. = 0.7844), low-density lipoprotein (LDL) (cor. = 0.7508), body mass index (BMI) (cor. = 0.6851) and 400-m walking pace (cor. = 0.7085). (Fig. 4b), suggesting subsets of SASP that are implicated in these clinical traits. A LASSO model also showed higher accuracy than a model using covariates alone (such as age, sex, race and kidney function) in predicting obesity (Fig. 4d). These models include covariates that contribute to the predictive potential of the model, but LSP-only models show remarkably similar correlation coefficients to LSP + covariate models (Extended Data Fig. 4a,b). This suggests that LSPs have clinical value independent of their age association. Further, to test the relative importance of inflammation

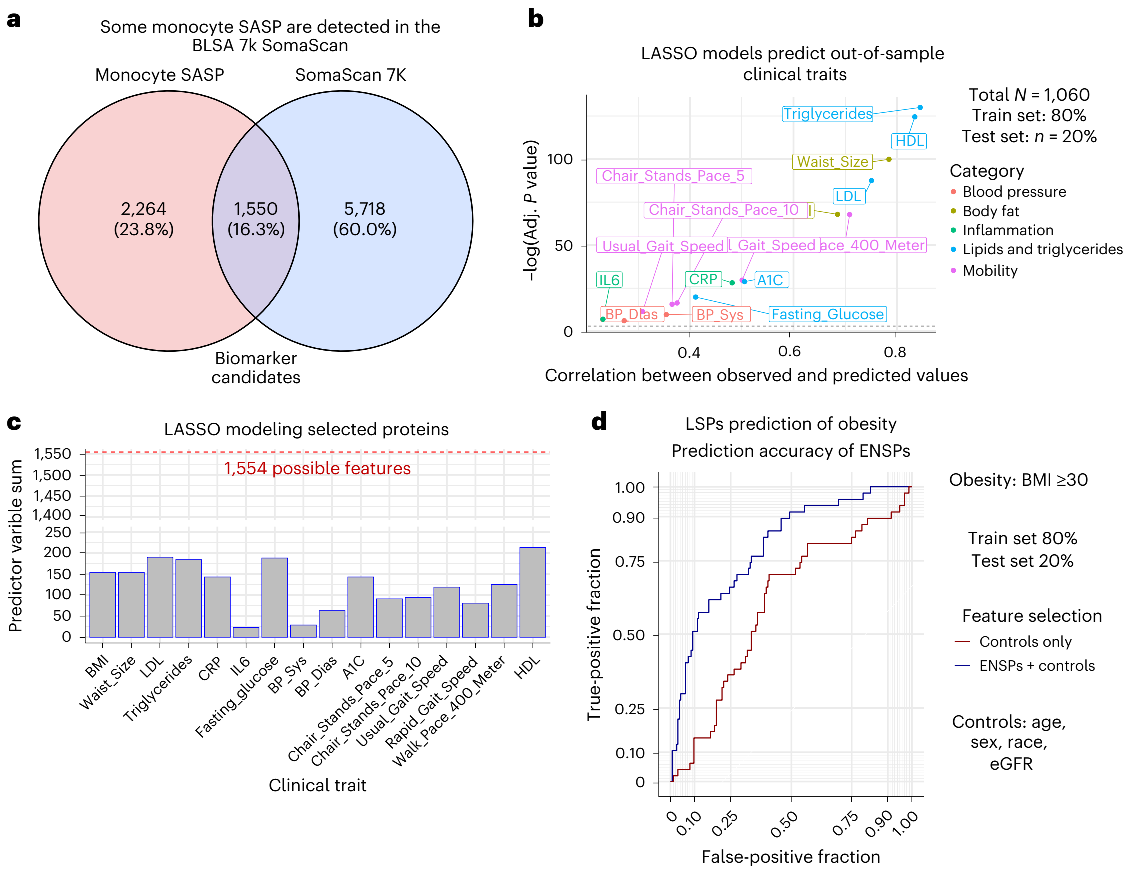

Fig. 4 | LASSO Modeling Using SASP of Clinical Traits in the BLSA. a, The 1,550 THP-1 monocyte SASPs are detected in the BLSA 7K SomaScan. **b**, LASSO models were trained on 80% of the BLSA cohort and used to predict clinical traits of the remaining 20%. Total sample sizes per trait are indicated in Table 1. Two-sided Spearman correlations are shown between the predicted and observed values in the test set for each clinical trait. **c**, LASSO modeling was used for feature selection and the number of LSPs implicated in each clinical trait are shown, including 1,550 SASP proteins and four covariates (age, sex, race and eGFR). **d**, /Receiver operating characteristic (ROC) plot comparing the predictive potential (80% train and 20% test) of LSPs to predict obesity with control-only models, showing that LSPs seem to provide additional predictive potential beyond age and other controls alone.

as a driver of the model associations, interleukin (IL)-6 and C-reactive protein (CRP) levels were also considered as a covariate during feature selection to build a separate model used to predict test set BMI and walking pace (Extended Data Fig. 5a). This analysis revealed that association with inflammation, as assessed by measuring CRP or IL-6 levels, is not a crucial component of the SASP models.

Association of SASP signatures with body fat depots

To explore whether the high predictive potential of LSPs for clinical traits related to BMI and waist size were driven by specific fat depots in the body, we performed LASSO modeling of whole-body computed tomography and dual X-ray absorptiometry scans of the BLSA participants, which were used to quantify body fat content at various regions across the body, including subcutaneous fat in the limbs and abdomen, visceral fat and intramuscular fat depots. A significant predictive potential was found across all body fat measures in the test dataset (Fig. 5a) when utilizing LSPs and covariates that were selected for each trait (Fig. 5b). Relatively high correlations between observed and predicted values were observed across different fat depots, ranging from 0.5074–0.7473, with the highest correlation in trunk fat and the lowest in abdominal subcutaneous fat. Moreover, total body fat percentage was most strongly predicted by models trained using senescence signatures and covariates (Spearman Cor. = 0.7913) than any individual fat depot (Fig. 5a,d). Because fat percentage is adjusted for body size, these data suggest that overall proportion of body fat, rather than the size of any specific fat depot, is associated with a THP-1 monocyte SASP. LSPs were also used to predict fat percent in a test cohort and showed higher accuracy than a covariate-only (age, sex, race and estimated glomerular filtration rate (eGFR)) model when predicting fat percent-based obesity (Fig. 5c), suggesting that LSPs have age and covariate independent predictive potential.

Association of SASP signatures with a composite frailty index

Senescence signatures were also examined for their association with frailty in the BLSA. Senescence signatures associated with the frailty index[24,25] were identified via LASSO modeling, and linear models using only the selected proteins demonstrated moderate cross-validated (90% train and 10% test) frailty prediction ability in the BLSA (Extended Data Fig. 5b). Additionally, among the selected senescence signatures implicated in frailty, linear modeling revealed that a subset was positively associated with frailty (Extended Data Fig. 5c).

Validation in an independent aging cohort

To further validate the role of THP-1 monocyte SASPs in multiple phenotypes. We performed a cross-validation analysis using plasma proteomic data from BLSA and InCHIANTI (demographics and clinical traits summarized in Supplementary Table 3), a population-based study of older individuals living in the Chianti geographic area of Italy, which was assessed using the 1.3K SomaScan assay, an earlier version of the assay containing a subset of the total protein measurements in BLSA. We identified 220 SASP candidates detected in both cohorts that were selected for further testing for clinical trait associations.

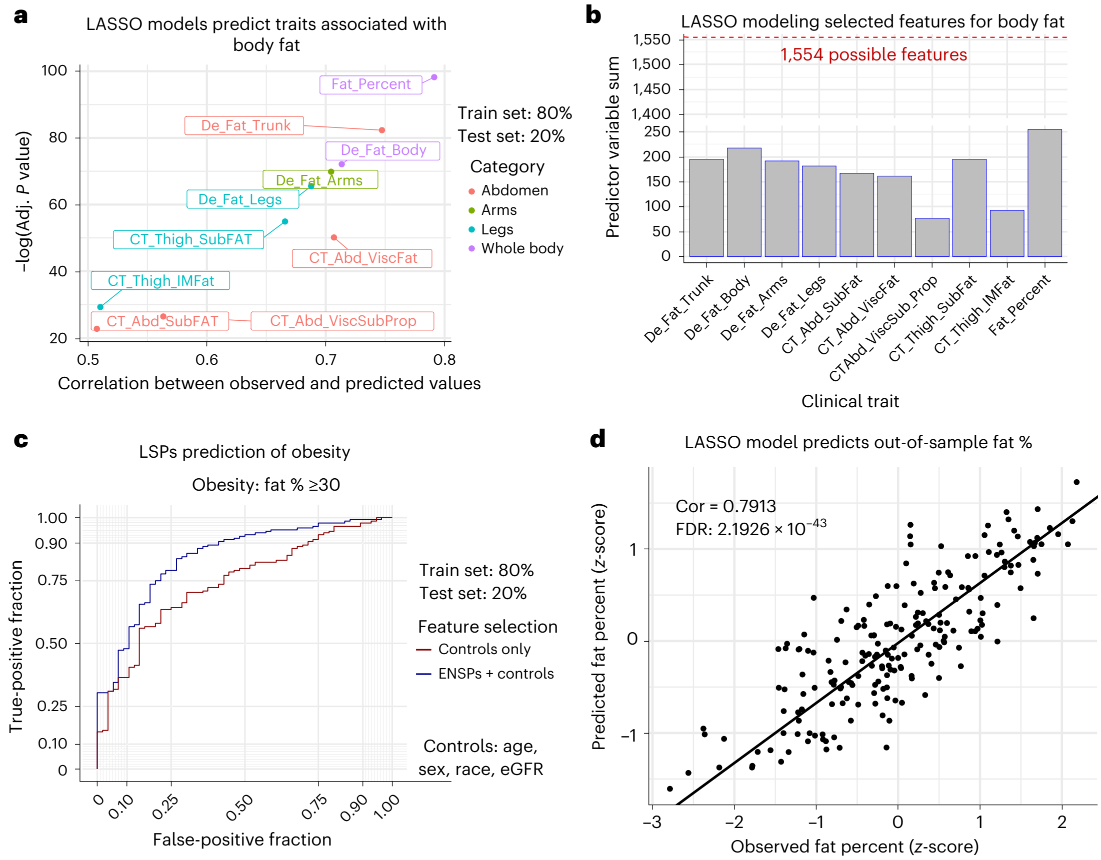

Fig. 5 | Modeling of the fat content in BLSA using SASP candidates. a, LASSO models were trained on 80% of the BLSA cohort and used to predict clinical traits of the remaining 20%. Total sample sizes per trait are indicated in Table 1. Two-sided Spearman correlations are shown between the predicted and observed values of the test set for each clinical trait. **b**, LASSO modeling was used for feature selection, and the number of LSPs implicated in each clinical trait are shown. **c**, ROC plot comparing the predictive potential (80% train and 20% test) of LSPs to predict obesity with control-only models, showing that LSPs seem to provide additional predictive potential beyond age and other controls alone. **d**, The correlation between observed waist size and that predicted by LASSO modeling (80% train and 20% test).

LSPs were identified for several clinical traits in both BLSA (Extended Data Fig. 6a) and InCHIANTI (Extended Data Fig. 6b). LSPs identified in both cohorts were used to train linear models used for cross-validation. These linear models, when trained on the BLSA cohort, significantly predicted several clinical traits in InCHIANTI (Extended Data Fig. 6c) and vice versa (Extended Data Fig. 6d). Overlapping LSPs effectively cross-validated in both directions for several clinical traits including BMI, triglycerides and walking pace, among others in BLSA and InCHIANTI (Fig. 6a), indicating a robust association of these proteins with clinical traits. Additionally, a binomial model trained on overlapping LSPs in BLSA predicted obesity in InCHIANTI better than an age and sex-only model (Fig. 6b). The cross-validation potential of LSPs for a subset of clinical traits, such as BMI, blood pressure, triglycerides and walking pace, in geographically and genetically distinct human cohorts demonstrates their robust clinical relevance.

SASP panel prediction of aging and obesity-related outcomes

To explore whether a substantially smaller subset of the SASP could be leveraged as a robust and clinically relevant panel across multiple clinical traits, we prioritized and selected a smaller set of proteins based on their associations with multiple traits and relative importance. Though a different subset of SASPs was selected for each trait, consistent with previous reports[11,26], notable proteins were selected via LASSO modeling in several traits of a 14-trait panel. Ranking the LSPs by the number of features in which they were implicated revealed that NQO1 and IL1RN were selected for 8 of the 14 traits and LMAN2, IGF2R, KHSRP and CCL18 were selected for 7 of the 14 traits. A total of 21 LSPs were identified in at least 5 of the 16 traits, 7 of which were also quantified in the InCHIANTI 1.3K panel (Fig. 7a). Linear models of the high-impact panel that were trained on 80% of the cohort significantly predicted many clinical traits of the remaining 20% test set in both BLSA (Fig. 7b) and InCHIANTI (Fig. 7c). Next, the high-impact panel expression levels were condensed into a single continuous variable using PCA. In this way, principal component 1 was used to represent each individual's senescence burden score. Ranking the BLSA and InCHIANTI cohorts by their composite senescence burden and plotting linear trends of the clinical traits walking pace, HDL, BMI and CRP reveals that increasing senescence burden score shows the expected association with trait trending in the direction of poorer health (positively associated with BMI and CRP, negatively associated with walking pace and HDL), in both BLSA (Fig. 7d) and InCHIANTI (Extended Data Fig. 7a). This suggests that the high-impact panel could potentially be used as a metric of individual senescence burden in a clinical setting.

Senescence signatures are age and covariate independent

To select senescence signatures that are implicated in clinical traits irrespective of age and other covariates, such as race, sex and kidney function (eGFR), these metrics were included in the LASSO modeling performed thus far and therefore contributed to the predictive potential of these models. To evaluate the contribution of covariates such as age to the predictive potential of LASSO models, we compared covariate-only models to combined LSP + covariate models by

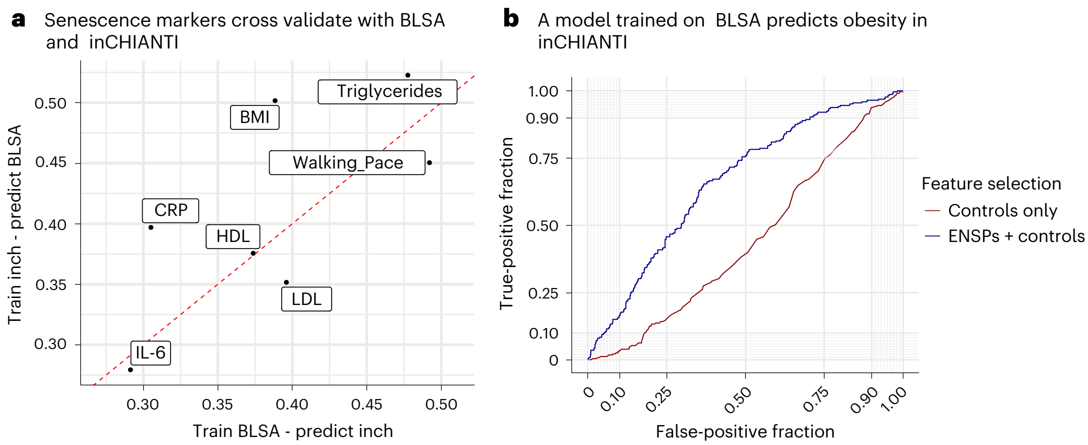

Fig. 6 | SASP-based associations show robust replication in the InCHIANTI aging study. a, 220 THP-1 monocyte SASP were detected in both the BLSA (7K SomaScan) and InCHIANTI (1.3K SomaScan). LASSO modeling was used for feature selection in both InCHIANTI ($n = 997$) and BLSA ($n = 1,060$) and linear models were constructed using only proteins selected in both studies for each trait. Spearman's correlation of predicted values of linear models trained on the BLSA and observed values in InCHIANTI are shown on the x axis and Spearman's correlation of predicted values of linear models trained on InCHIANTI and observed values in the BLSA are shown on the y axis. **b**, Binomial models were trained either using controls (age and sex) or controls + LSPs in BLSA, then used to predict obesity in InCHIANTI.

testing the analysis of variance (ANOVA). ANOVA testing revealed that including LSPs in linear models significantly improved the accuracy of covariate-only linear models (Supplementary Table 4) and substantially increased correlation coefficients (Extended Data Fig. 4a) and LSP-only models produced similar correlation coefficients to LSP with covariates models (Extended Data Fig. 4a). A similar analysis was conducted using whole-body fat percent as an additional covariate along with age, sex, race and eGFR to determine whether LSPs show body fat independent predictive potential. ANOVA testing again revealed that LSPs often added significant additional predictive accuracy to models using only body fat and other covariates (Supplementary Table 4), and adding LSPs to body fat percent with other covariate-only linear models again increased the correlation coefficient for many clinical traits, particularly triglycerides, HDL, LDL, CRP and fasting glucose (Extended Data Fig. 4b). Additionally, for each clinical trait, the predictive potential of LSPs was compared to a randomly selected group of proteins of the same group size. Permutation tests (100,000 repetitions per trait) revealed that LSPs often demonstrated better out-of-sample trait prediction potential than randomly selected proteins (Extended Data Fig. 7b and Supplementary Table 5). These results implicate senescence-associated proteins associate with BMI and other clinical traits independently of the effects of aging and other covariates.

Discussion

This study applied a nanoparticle-based MS strategy to identify SASPs from THP-1 cells in fully supplemented culture conditions and revealed circulating senescence signatures that predict aging-associated clinical traits in humans. MS-based proteomics allows a comprehensive unbiased characterization of the cell secretome, yet the large numbers of proteins, such as albumin in FBS, in most culture media are a major hurdle in MS-based analysis of SASP proteins. Our group and others addressed this issue by using serum-free medium[15,17,27–30]; however, prolonged absence of serum profoundly alters cellular phenotypes, metabolism and viability. Serum starvation can also trigger inhibition of mTORC1; mTOR is a potent regulator of the SASP in cultured cells[31,32]. Application of the automated, nanoparticle-based workflow here enabled the comprehensive profiling of the SASP in THP-1 cells under fully supplemented culture conditions and free of the confounders of starvation.

Our workflow adapted a recent technology that has been used to aid in comprehensive detection of proteins in samples with a large dynamic range of protein concentrations[16,20,21,33]. In this approach, the protein mixture bound at the interface of the surface of nanoparticles and a protein sample, termed the 'protein corona', contains a greatly reduced protein dynamic range and allows the detection and quantification of proteins that are normally undetectable in blood[16,20,21,34]. Because serum supplementation essentially produces the same dynamic range problem as conditioned medium, we reasoned that the same workflow would enable the comprehensive, quantitative and unbiased profiling of the SASP under fully supplemented culture conditions. Indeed, we detected a dramatically increased number of human peptides in conditioned medium supplemented with FBS.

Inflammatory chemokines, interleukins and growth factors are among the most widely cited bioactive components of the SASP[35] and commonly used clinical indicators of inflammation, namely IL-6, are themselves widely known SASP factors. Thus, associations between SASP and inflammation are expected. In the present study, we note that SASP models were associated with inflammation despite the absence of either IL-6 or CRP among the protein panels. Similarly, it is important to note that it is difficult to separate the effects of senescence-associated inflammation from other sources of inflammation as drivers of associations noted in this study. Besides inflammatory components, the SASP also includes a number of other signaling factors such as growth factors (such as GDF15 and IGFBPs), proteases and their inhibitors (such as MMPs and SERPINs), extracellular matrix (ECM) components, and other factors characteristic of the senescence proteome[15]. Notably, both the full SASP models and the high-impact panel from this study incorporate a diverse array of SASP components that include both inflammatory and non-inflammatory components and likely capture the effects of both. We note that the major associations captured by this study are related to body composition and metabolic parameters, whereas the associations with inflammation (IL-6 or CRP levels) are relatively modest by comparison. Further, the inclusion of inflammation as a covariate has a minor effect on the strength of the model (Extended Data Fig. 5a), suggesting that other components of SASP are also important biomarkers of these phenotypes.

Our results are consistent with the premise that senescent cells, particularly monocytes, may either contribute to or be driven by decline in diverse age-related and obesity-related clinical outcomes, including loss of mobility, increased body fat (BMI, fat percentage and waist size), increased blood pressure, elevated triglycerides and lipids, elevated glucose and glycated hemoglobin (HbA1c) and inflammation (CRP and IL-6). These findings are also consistent with previous

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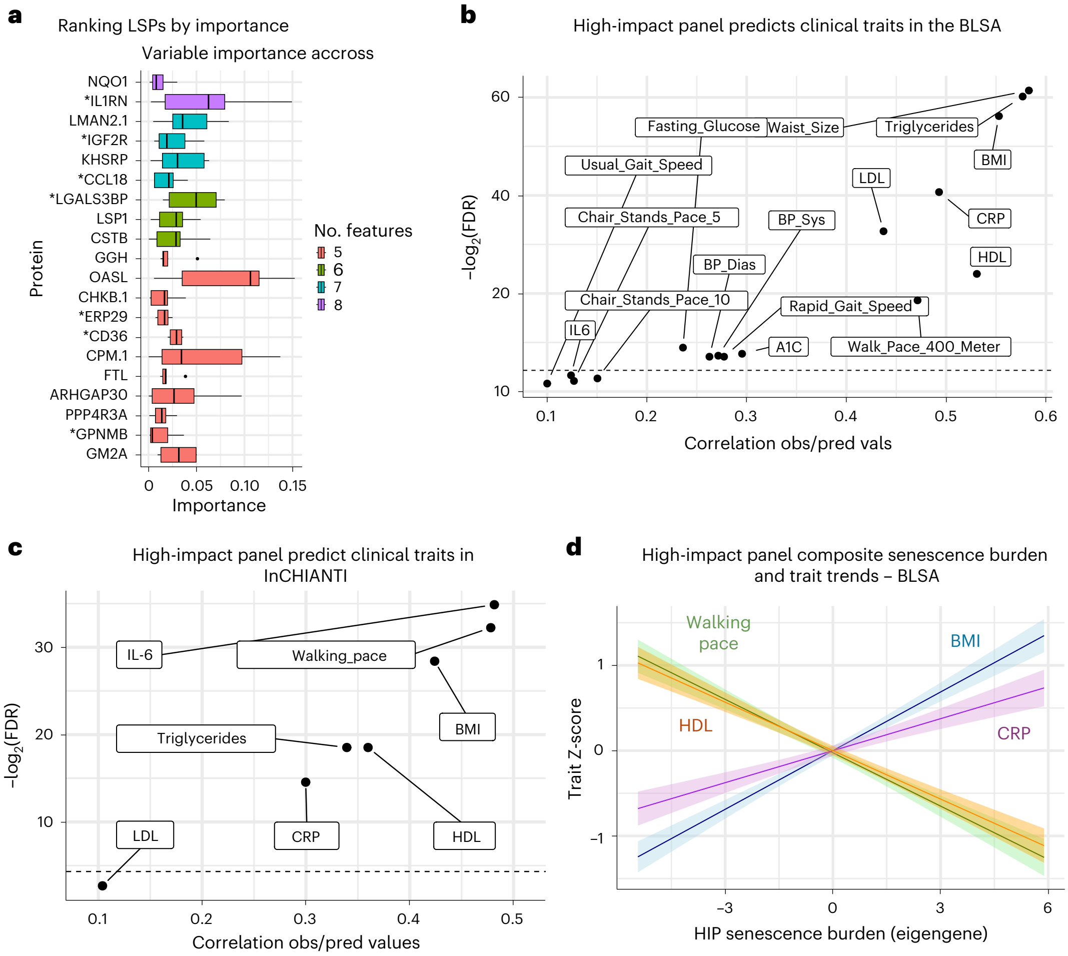

Fig. 7 | A high-impact SASP panel robustly predicts multiple clinical traits. **a**, For a 14-trait panel, proteins were ranked by the number of features for which they were selected via LASSO in the BLSA, and the most frequently selected proteins are shown with their cross-trait importance on the *x* axis. Only proteins that were positively associated with negative traits such as BMI and CRP, and those that were inversely associated with positive traits such as mobility were selected. Stars indicate those that were also detected in InCHIANTI. Boxplot elements are defined as follows: center line, median; box limits, upper and lower quartiles; whiskers, 1.5× interquartile range; points, outliers. **b**, Linear models were trained on 80% of the BLSA cohort and used to predict clinical traits in the remaining 20%. Total sample sizes per trait are indicated in Table 1. Spearman correlation between the predicted and observed test values are shown. **c**, Linear models were trained on 80% of the InCHIANTI cohort and used to predict clinical traits in the remaining 20%. Spearman correlation between the predicted and observed test values are shown. **d**, PCA was used to condense the high-impact panel into a composite senescence burden score in the BLSA. Principal component 1 was used to represent an eigengene for the high-impact panel. With the BLSA cohort ranked from a low to moderate to a high senescence burden, linear trait trends reveal that the positive traits HDL and walking pace show a negative trend, whereas the negative traits BMI and CRP show a positive trend. The lines represent the trait scores predicted by each linear model. Shaded regions represent 95% confidence intervals.

studies in multiple aging cohorts identifying senescence markers that are associated with diverse clinical traits of aging, such as frailty and cognitive decline. For example, SASP proteins, such as ICAM1, MMP7 and Activin A, have been associated with a decline in physical activity in participants of the LIFE study[9]. In addition, plasma levels of 13 core SASP proteins, including CTSB, a component of the THP-1 monocyte SASP, are associated with all-cause mortality and multimorbidity in the BLSA[11]. Few of the published SASP-derived protein associations were observed in the present study, highlighting the heterogeneity in senescent cell phenotypes based on cell type. For example, GDF15 is notably missing from the monocyte SASP, while the most important SASP proteins (based on the number of associations, Fig. 7a) identified in the present study are not among the circulating senescence factors summarized in current literature[7].

One of the notable findings of this study is the robust association between the THP-1 monocyte SASP and obesity-related outcomes, such as BMI and body fat, which were among the strongest associations observed. Body fat percentage, a measure of body fat corrected to overall body size, was more strongly associated with THP-1 monocyte SASP than any individual fat depot, suggesting that fat proportion, rather than overall mass, is likely associated with senescence. The association of THP-1 monocyte SASP and body fat and related outcomes is notably independent of age and other covariates (Extended Data Fig. 4a and Supplementary Table 4), suggesting that senescent cells may contribute to these pathologies independent of aging. These results suggest a potential link between monocyte senescence and obesity.

While the direction of causality cannot be definitively determined in the present study, there is ample evidence to support the obesity as a driver of cellular senescence. Culture conditions that mimic aspects of obesity, such as high free-fatty acids or glucose, can drive cells into senescence in vitro[36–38]. Senescent cells accumulate in obesity and high-fat diet, particularly in adipose tissue[39,40]. Furthermore, the transplantation of senescent preadipocytes into mice fed a high-fat diet exacerbates declines in walking speed and endurance when compared to normal diet[41]. Thus, evidence strongly points to obesity as a disease-associated senescence inducer that can be decoupled from aging. Additionally, results from the CALERIE trial demonstrate the reduction of senescence biomarkers in individuals over 1–2 years of calorie restriction, further suggesting a link between senescence burden and diet, body composition and metabolism[42]. We speculate that at least some of the clinical traits described in this study can be attributed to obesity-associated senescent cells. Indeed, we observe that SASP-based LASSO models predict key obesity-associated clinical outcomes related to metabolism (fasting glucose and hbA1c), lipids (triglycerides and cholesterol) and blood pressure. Collectively these findings suggest the importance of considering obesity as a key contributor to senescence and senescence-associated outcomes in humans and suggest that individuals with obesity may be among those that benefit most from serotherapeutic interventions in future trials.

As a hallmark of aging, it is important to explore whether the association of SASP with clinical phenotypes can be uncoupled from aging to some extent. We indeed observed that predictive models based on SASP added value to models that include covariates (age, sex, race and eGFR) and were clinically meaningful in predicting outcomes such as obesity. SASP models showed significant cross-study prediction ability, notably for several obesity-related traits including BMI, cholesterol and triglycerides (Fig. 6a), despite the reduced number of features available for the validation study (limited to proteins detected in both BLSA and InCHIANTI). Moreover, a small panel of high-impact senescence proteins associated robustly with multiple traits in both cohorts (Fig. 7), supporting the clinical significance of these SASP factor subsets when applied selectively, and mitigating potential concerns of overfitting posed by models based on the full SASP.

A clinically meaningful finding was that a relatively small panel of SASP robustly predicted a set of age- and obesity-associated clinical outcomes, including inflammation (IL-6 and CRP), lipids (HDL and LDL), glucose (hbA1c and fasting glucose), blood pressure, walking speed and pace, and BMI. Several proteins in the panel are consistent with senescence biology and have known associations with outcomes with elevated senescent cell burden. Notably, the cytokine CCL18 (also known as PARC) previously showed the strongest association with mortality among 28 SASPs in a study of 1,923 individuals over the age of 65 years (ref. 43) and has been associated with disease-progression or negative outcomes in a range of diseases including cancer[44] and atherosclerosis[45]. In future studies, it will be valuable to validate the associations of the set of proteins in the high-impact senescence panel with elevated senescence burden and evaluate their potential roles as either drivers or biomarkers of disease outcomes.

This study has several limitations. SASP factors in plasma can be contributed by a variety of cells and tissues in the body. Thus, it is not possible to track the originating tissues of SASPs in circulation or to verify whether circulating proteins were released by senescent cells or other secreting cells with common secretory factors, such as activated immune cells. Senescence signatures are numerous and heterogeneous by cell type[15,46], and examining clinical associations of senescence signatures from a variety of tissue types (senotype-specific) is warranted. The cellular model used in this study, THP-1, is an immortalized cell line derived from peripheral blood of a childhood case of acute monocytic leukemia that resembles primary monocyte cultures in morphology and differentiation properties[47,48]. It is noteworthy, however, that the mechanistic observations made in these cells should be interpreted with caution and validated in primary monocytes and in animal models. Finally, longitudinal proteomic measurements will be useful in future studies for evaluating whether SASP protein trajectories can be more sensitive in predicting clinical outcomes.

In summary, we showed that SASP factors from THP-1 monocytes have a significant association with aging-associated clinical traits and can serve as biomarkers to predict biological aging. Using nanoparticle-based enrichment coupled with MS enabled comprehensive characterization of the secretome from senescent THP-1 monocytes in culture in serum-supplemented culture conditions. Our results highlight an approach to study the cellular secretome under physiological conditions. Moreover, this study sheds light on clinical associations of circulating SASPs in human longitudinal studies and identifies possible biomarkers of senescence that could potentially inform future senotherapeutic trials in aged individuals and individuals with obesity.

Methods

Ethics statement/regulatory approvals

All human studies were compliant with ethical regulations. The BLSA protocol (03AG0325) was approved by the institutional review board of the National Institute of Environmental Health Science, part of the National Institutes of Health (NIH). The InCHIANTI study (exemption no. 11976) protocol was approved by Medstar Research Institute (Baltimore, Maryland), the Italian National Institute of Research and Care of Aging Institutional Review, and the Internal Review Board of the National Institute for Environmental Health Sciences. All participants in BLSA and InCHIANTI provided written informed consent and were not compensated.

Cell culture and senescence induction

THP-1 human monocytes (ATCC, TIB-202) were cultured in RPMI 1640 medium (Thermo Fisher Scientific, 11875119) supplemented with 10% FBS (Thermo Fisher Scientific, 26140079) and 1% pen–strep (Thermo Fisher Scientific, 15070063) in a 20% O_2, 5% CO_2 incubator. To determine the optimum conditions for induction of senescence, proliferating THP-1 cells were exposed to different doses of ionizing (γ) radiation (IR; 5, 7.5 and 10 Gy) using a Gammacell 40 Exactor, a Cesium-137 based irradiator (Nordion); thereafter, cell viability and senescence markers were assessed over different time points. Cell viability was measured using the CellTiter-Glo Luminescent Cell Viability Assay (Promega Corporation, G7570) as per the manufacturer's instructions. IR radiation of 7.5 Gy and culture up to 7 days were optimum for senescence induction and were used for all further experiments. Proliferating cells were used as nonsenescent controls for all experiments.

EdU incorporation assay

Cell proliferation was assessed using the Click-iT Plus EdU Cell Proliferation kit (Thermo Fisher Scientific, C10640), following the manufacturer's instructions. In brief, 20,000 cells per well were seeded in 100 µl of medium in 96-well plates in four replicates. At 6 days after irradiation, cells were incubated with 20 µM 5-ethynyl-2′-deoxyuridine (EdU) overnight at 37 °C. The next day, cells were fixed using 3.7% formaldehyde for 15 min, followed by permeabilization with 0.5% Triton X-100 for 20 min. Cells were then incubated with Alexa Fluor picolyl azide and Hoechst 33342 dye for 30 min each. The resulting images were captured by a fluorescence microscope (BZ-X Analyzer, Keyence Corporation). The percentage of cells that showed EdU incorporation corresponded to the percentage proliferating cells.

Quantitative PCR analysis

Total RNA was isolated from proliferating and senescent THP-1 cells using a Direct-zol RNA Miniprep Kit (Zymo Research, R2052) and 500 ng of total RNA was reverse transcribed using Maxima reverse transcriptase (Thermo Fisher Scientific, EP0741) and random hexamers. For reverse transcription followed by quantitative PCR (RT–qPCR) analysis, 0.1 µl cDNA was employed with 250 nM of gene-specific primers (primer sequences provided in Supplementary Table 6) and KAPA SYBR FAST qPCR kits (KAPA Biosystems) on the QuantStudio 3 Realtime PCR system (Thermo Fisher Scientific). Relative RNA levels were calculated after normalizing to *ACTB* mRNA encoding the housekeeping protein β-actin using the $2^{-\Delta\Delta Ct}$ method.

Senescence-associated β-galactosidase assay

SA-β-gal activity was assessed using the Cellular Senescence Plate Assay kit (SPiDER-βGal; Dojindo Molecular Technologies, SG05), following the manufacturer's instructions. In brief, 15,000 cells of proliferating and senescent THP-1 cells in three replicates each were seeded in a 96-well plate. At 7 days after irradiation, cells were lysed and incubated with SPiDER-βGal for 1 h at 37 °C, and SPiDER-βGal fluorescence intensity was measured at 535 nm excitation and 580 nm emission using a microplate reader. For conventional SA-β-gal assay, wells of a 24-well plate were coated with Cell-Tak cell adhesive (Corning, 354240) and 150,000 cells were seeded in serum-free medium in three replicates. SA-β-gal activity was then assessed using a Senescence Detection kit (Abcam, ab65351) as per the manufacturer's instructions. In brief, cells were fixed and incubated with X-gal, a substrate of β-galactosidase enzyme, for 72 h at 37 °C. X-gal is hydrolyzed by β-galactosidase to form a product that forms blue coloration after oxidation. Images were captured using brightfield microscopy on a Rebel microscope (Echo) The number of cells forming the blue color was counted to determine the percentage of cells positive for SA-β-gal.

Flow cytometry

Proliferating and senescent THP-1 cells were collected 7 days after irradiation, washed with cold PBS and 2×10^6 cells were resuspended in 200 µl cold stain buffer (BD Biosciences, 554656). To identify dead cells, the cell suspension was incubated with 7-amino-actinomycin D (BD Biosciences, 420403) for 10 min in the dark at room temperature. 7-Amino-actinomycin D is a nucleic acid dye that permeates the membranes of dead cells and binds to DNA, allowing their discrimination from live cells. The cells were then stained with the following fluorochrome-conjugated antibodies by incubating in the dark at room temperature for 15 min: anti-human CD14-PE (BioLegend, 325606, dilution 1:40) and anti-human CD11b-BV605 (BioLegend, 301331, dilution 1:40). Cells were then washed twice with cold stain buffer, centrifuged at 1,000*g* at room temperature for 10 min, and resuspended in 300 µl stain buffer. Flow cytometry was performed on a Cytoflex LX flow cytometer (Beckman Coulter) by collecting 10,000 events for each sample and the data were analyzed using FlowJo software v.10.7.1 (Becton Dickison & Company) and the expression on CD14 and CD11b was determined within the live cell population.

Sample preparation for mass spectrometry

SASPs from senescent and proliferating THP-1 monocytes were collected, based on a modified version of the protocol from Neri et al.[17], adjusting the protocol so that complete medium (supplemented with 10% FBS) was used in place of serum-free medium. In brief, THP-1 cells were grown in T75 flasks to sub-confluence before induction of senescence. For senescent samples, cells were shifted to fresh complete medium 7 days after irradiation, and conditioned medium containing SASP was collected after 48 h. Conditioned medium from proliferating cells was similarly collected after a 48-h incubation in fresh complete medium. Conditioned medium was then centrifuged at 10,000*g* for 15 min to pellet down cell debris. One milliliter of each conditioned medium sample was then aliquoted into tubes for downstream preparation, for a total of 14 samples (seven senescent and seven nonsenescent).

The conditioned medium samples were processed as described[16] on the SP100 Automation Instrument coupled with Proteograph XT Assay kits (Seer). In brief, samples were incubated with two chemically distinct nanoparticle suspensions supplied in the assay kits. Nanoparticle-bound proteins were captured by incubation and a series of gentle washes. The resulting protein coronas were reduced, alkylated, and digested with Trypsin/Lys-C to generate peptides for MS analysis. The peptide mixture was desalted using solid phase-extraction. All steps in the preparation of peptides were conducted automatically by the SP100, which produces two peptide fractions per sample, one fraction for each nanoparticle suspension.

In addition to the preparation of 14 nanoparticle-processed samples, six matched neat samples were prepared (three senescent and three nonsenescent). Neat samples were processed in parallel with the other samples, except that no nanoparticles were added and no protein corona were captured. Essentially, neat samples are equivalent to a standard digestion protocol and are used for comparison against standard methods. After processing samples with the Proteograph XT workflow, 34 peptide samples were generated for downstream MS analysis: 28 experimental samples (14 replicates with two nanoparticle fractions each) and 6 matched neat samples.

LC–MS

All samples were analyzed using a Vanquish Neo UHPLC system coupled with an Orbitrap Astral mass spectrometer (Thermo Fisher Scientific) with a NanoSpray Flex source (Thermo Fisher Scientific). Peptides (400 ng of each sample) were loaded on Acclaim PepMap 100 C18 (0.3 mm ID × 5 mm) trap column and then separated on a 50-cm µPAC analytical column (PharmaFluidics, Belgium) at a flow rate of 1.0 µl min^{-1} using a gradient of 4–35% solvent B (0.1% formic acid in acetonitrile) mixed into solvent A (0.1% formic acid in water) over 20.8 min and a total run time of 24 min. The MS data were acquired in DIA mode with a normalized HCD-collision energy of 25% and a default charge state of +2. MS1 spectra were acquired in the Orbitrap every 0.6 s at a resolving power of 240,000 at m/z 200 over m/z 380–980. The MS1-normalized AGC target was 500% (5×10^6 charges) with a MaxIT of 5 ms. For MS/MS experiments, the DIA experiment was set at a 3-m/z isolation window, making 199 DIA scan events across the precursor isolation windows spanning 380–980. MS2 scans were collected from 150–2,000 m/z. DIA MS2 scans were acquired in an Astral analyzer with a normalized AGC target of 500% (5×10^4 charges) to better control the ion population using MaxIT that was set to 5 ms. Window placement optimization was turned on. The MS acquisition method had a mean of 8.002 data points per peak (median of 8). A source voltage of 1,500 V and an ion-transfer tube temperature of 280 °C were used for all experiments.

Protein identification and quantitative analysis pipeline

Data were analyzed using the Proteograph Analysis Suite and the DIA-NN v.1.8.1 algorithm to perform the peptide identification and quantification using an in silico-predicted library, based on a combined database containing both UniProt human and bovine proteomes (from UniProt release 2022_04, released on October 11, 2022, containing 105,533 entries including isoform and TrEMBL). Match between runs was enabled. Otherwise, all DIA-NN settings were set to default. The FDR was set to 1% for filtering identifications at the peptide and protein group levels. Quantification of peak areas was performed at the MS2 level. The MS acquisition method had a mean of 8.002 data points per peak, ranging from minimum 4 (DIA-NN specific setting) and up to 18 data points per peak. Peptide quantification was performed using a max representation approach, where the single quantification value for a particular peptide represents the quantitation value of the nanoparticle most frequently measured across all samples, and standard peptide-to-protein rollup was used on selected peptide values. Furthermore, to specifically identify the THP-1 monocyte SASP and eliminate proteins originating from the FBS and unambiguously remove nonhuman peptides, all the peptides that mapped to the bovine proteome, or both human and bovine proteins, were removed from further analysis. Scaled intensities of human-unique peptides were aggregated to generate protein intensities using the MaxLFQ algorithm, implemented in the R/Bioconductor package iq (v.1.9.12)[49]. Differential analysis was conducted on the protein intensities between PRO and SEN samples using the R/Bioconductor package limma (v.3.58.1)[50].

Pathway analysis

Gene Ontology analysis was performed to identify the biological processes and molecular functions enriched among the differentially expressed proteins using ClusterProfiler (v.4.6.0)[51,52] in Rstudio 2023.06.0-421 (RStudio) and R v.4.2.0 (R Development Core Team). A background list of proteins was utilized in each analysis. In Fig. 3e, the background list included all human proteins identified by MS. In

Fig. 3g, the background list consisted of all proteins identified in both the fibroblast and THP-1 secretomes. In Fig. 4c, the background list included all proteins that overlapped between the SomaScan 7K assay and the THP-1 secretome proteins.

Clinical data

We used clinical data from the BLSA and InCHIANTI cohorts to determine the association of senescence markers with physiological markers of health and aging. The BLSA study is a population-based study that started in 1958 to evaluate the contributors of healthy aging in participants recruited from the DC/Baltimore metropolitan area who are at least 20 years old[53–55]. It involves data collection of clinical parameters, such as waist circumference, BMI and blood pressure, which are assessed during a standard medical exam. Blood tests were performed at a Clinical Laboratory Improvement Amendments certified clinical laboratory at Medstar Harbor Hospital, Baltimore. Total cholesterol was measured using alkaline phosphatase, HDL and LDL with dextran magnetic beads, triglycerides with colorimetric methods and glucose with glucose oxidase using the Vitros system (Ortho Clinical Diagnostics). Serum inflammatory markers IL-6 (R&D System) and CRP (Alpco) were measured by ELISA. The hbA1c levels were measured using LC by an automated DiaSTAT analyzer (Bio-Rad). Grip strength was measured three times on each of the right and left hand and the highest average grip strength was reported. Usual gait speed was measured in two trials of a 6-m walk and the faster time between the two trials is used for analysis. Body fat measurements were made using CT/dual X-ray absorptiometry scans. To avoid possible confounding effects of medication use, patients taking diabetes or hypertension medication had their hbA1c or blood pressure values excluded from statistical analysis.

The InCHIANTI is a similar population-based study of aging conducted in the Chianti region of Tuscany, Italy previously described in more detail[56]. Residents from the population registry of Greve in Chianti (a rural area) and Bagno a Ripoli (near Florence) ranging in age from 21–102 years participated in the study.

Frailty index

A composite frailty score was created as described previously in BLSA[25]. Frailty index procedures followed those established by Searle et al.[24] and included 44 common measures of frailty, with the addition of diabetes. Aspects of the composite frailty score include: reported difficulty carrying out 15 basic life tasks, including walking up ten steps and lifting 10 pounds; self-rated health from a health status survey; five items indicating emotional state from the Center for Epidemiologic Studies Depression Scale, including feeling lonely or depressed; four items indicating cognitive status from the mini-mental state examination, including orientation to time and memory; 14 binary health conditions including cancer and cognitive impairment, among others; recent weight loss, low physical activity, walking slowness and grip weakness.

SomaScan assay

The BLSA plasma proteomic study involved profiling for 7,596 SOMAmers using the 7K SomaScan Assay, whereas the InCHIANTI involved profiling for 1,322 SOMAmers using the 1.3K SomaScan Assay at the Trans-NIH Center for Human Immunology and Autoimmunity and Inflammation, National Institute of Allergy and Infectious Disease, NIH. Proteomic assessment and data normalization for each of the assays were conducted as reported[57]. Plasma protein concentrations were directly proportional to the abundance of the SOMAmer reagents that were reported in relative fluorescence units. The reliability and variability of the SomaScan assay measurements have been previously rigorously evaluated and applied in the BLSA[22,58].

Least absolute shrinkage and selection operator modeling

To avoid overfitting simple linear models, LASSO modeling, a penalized linear regression machine-learning technique, was used for feature selection of the most biologically relevant proteins. LASSO modeling was used due to its ability to perform unbiased feature selection, and because it can cope with high collinearity of features. The package glmnet[59] (v.4.1-6) randomly selected subsets of the dataset for cross-validation to find the lambda value with the smallest mean squared error. Due to its feature tuning method, this package produces slightly variable results. To account for this and to reduce variability, the package was run 100 times for each trait, and the most accurate penalty term was found across all runs.

Statistics and reproducibility

Unless otherwise specified, quantitative data have been represented as bar graphs with means ± s.d. Statistical analyses were conducted using unpaired two-tailed Student's t-test using GraphPad Prism v.9 and adjusted *P* values for multiple comparisons were calculated using Benjamini–Hochberg method. *P* values <0.05 were considered statistically significant. Unless otherwise specified, the linear models were generated using a two-sided Spearman correlation. The sample sizes in this study varied depending on the study and phenotype measured. Sample sizes ranged from 651 (DEXA measurements in BLSA) to 1,059 (BMI measurements in BLSA), with most measurements made in more than 900 individuals.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All raw MS data files and associated quantitative and statistical reports, metadata and supplementary data are available on MassIVE (dataset identifier: MSV000095315). FTP download link: ftp://massive.ucsd.edu/v08/MSV000095315/. The aggregated phenotype data have been provided as source data as well as supplementary tables. The non-aggregated clinical and SomaScan data are private because the participants did not consent to unrestricted data sharing at the time of the study conducted for BLSA. To comply with patient consent and data-sharing agreements, researchers are welcome and encouraged to request use of more detailed BLSA data for scientific projects by developing a pre-analysis plan that can be submitted for approval (https://blsa.nia.nih.gov/how-apply).

Code availability

The R scripts for the core LASSO analysis described[60,61] are available at https://github.com/geroproteomics/EN_Repeat/ and https://github.com/geroproteomics/EN_Test.

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Author contributions

R.B., D.T., A.R., T.N., M.S. and L.C. performed the experiments. B.O. performed the computational analysis. A.D. was involved in MS data analysis. R.B., B.O. and N.B. prepared the manuscript. T.T., G.D., Z.P. and J.C. helped in performing clinical associations. E.S., K.W. and L.F. provided the clinical data. M.G. and N.B. provided the facilities and guidance for the study.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s43587-025-00877-3.

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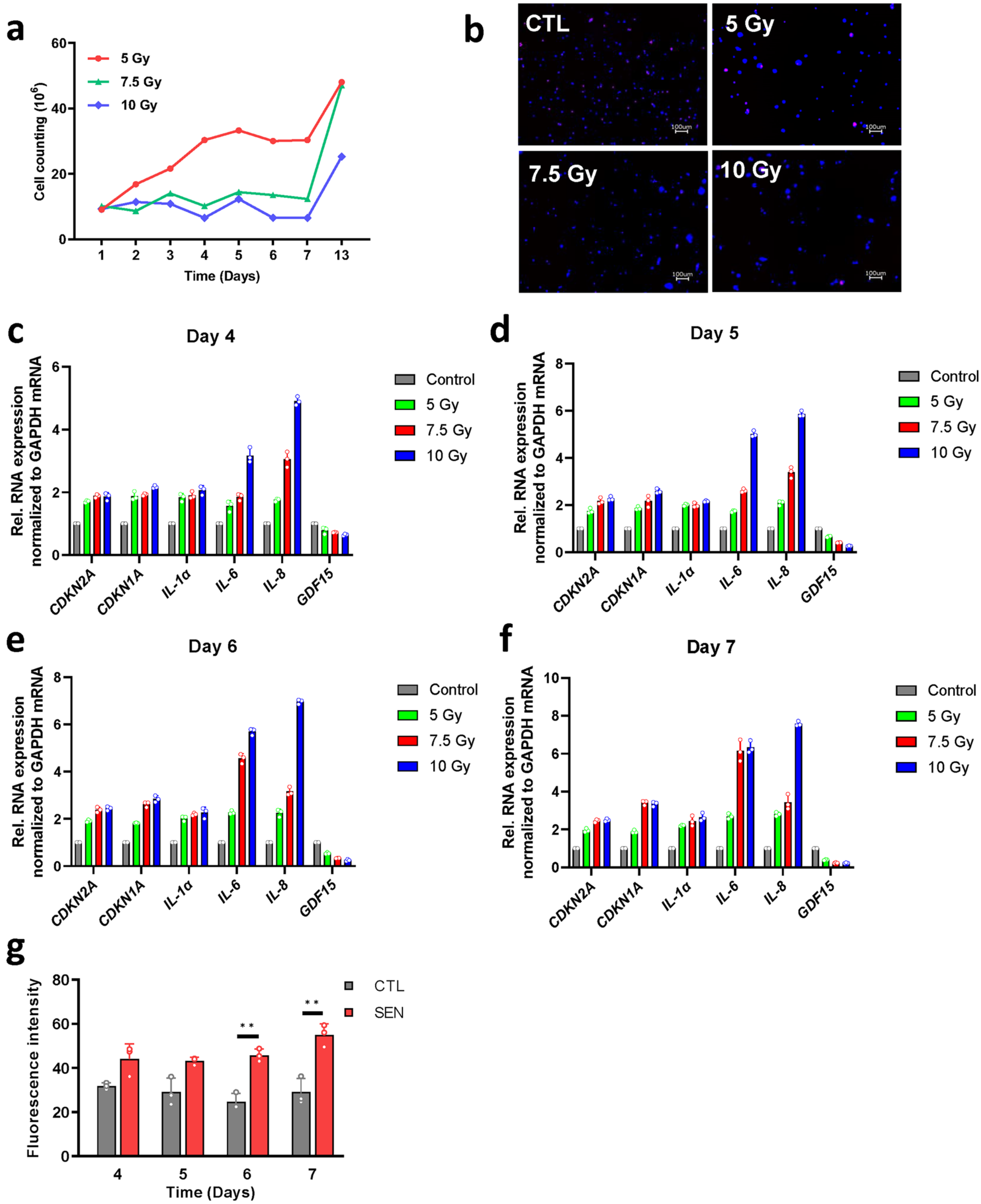

Extended Data Fig. 1 | Optimization of IR-induced senescence in THP-1 monocytes. a, Line plot indicating the cell number after exposure to different doses of IR indicates inhibition of cell proliferation up to 7 days after exposure to different doses of IR. **b**–**e**, Bar plots showing increased expression of known senescence markers over time after exposure to different doses of IR. **f**, Bar plots showing elevated SPiDER β-gal confirming induction of senescence in IR-treated THP-1 cells. Statistical analysis was performed using two-tailed Students' t-test (**$P < 0.01$). **g**, Representative fluorescence microscopy images from Edu incorporation assay after exposure to different doses of IR indicating inhibition of cell proliferation on Day 7 after exposure to different doses of IR. Data are represented as mean +/− standard deviation for n number of replicates in all graphs.

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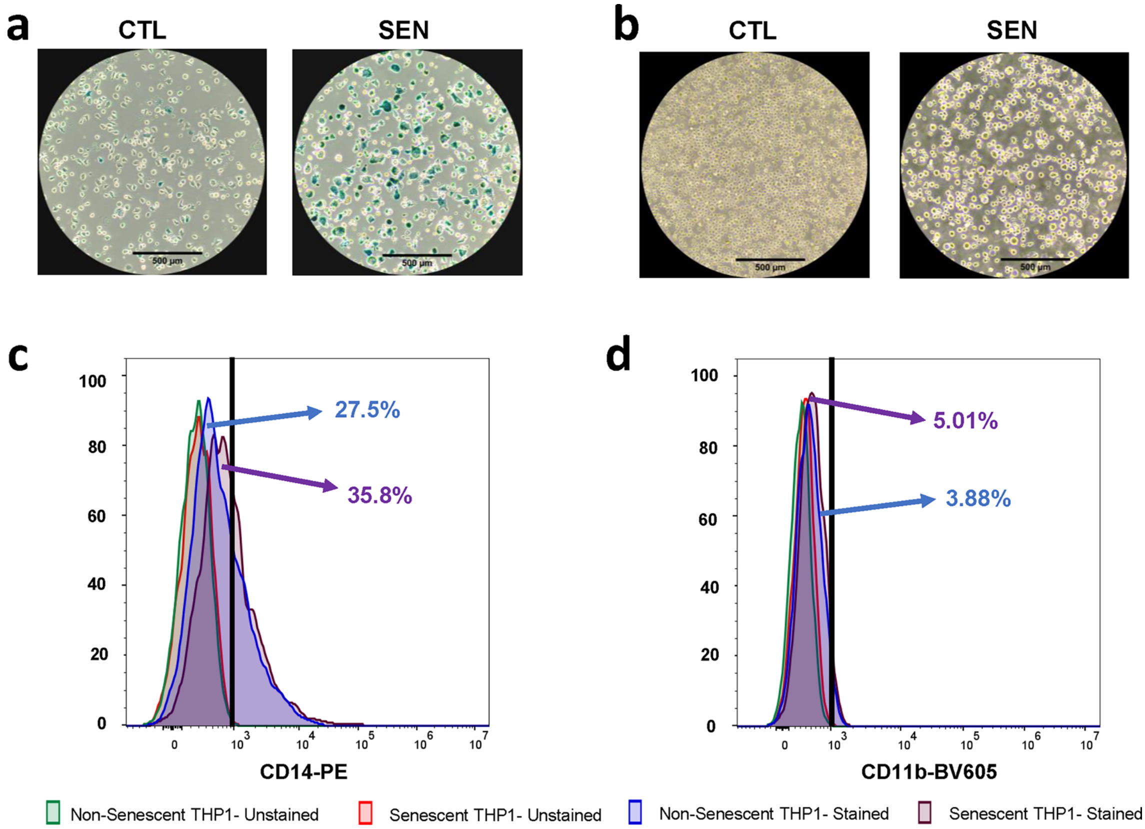

Extended Data Fig. 2 | Validation of Cell Identity after IR Treatment. a, Classical SA Beta-gal staining performed 7 days after IR treatment showed increased number of beta galactosidase positive cells as indicated by the formation of blue color in senescent cells as compared to the proliferating controls. **b**, Brightfield microscopy images to assess any change in morphology after senescence induction shows conservation of suspension cell characteristic of monocytes 7 days after IR treatment. Flow cytometry analysis performed to confirm cell identity after IR treatment shows no increase the monocyte differentiation markers **c**, CD14 and **d**, CD11b indicating conservation of monocyte cell lineage after IR treatment.

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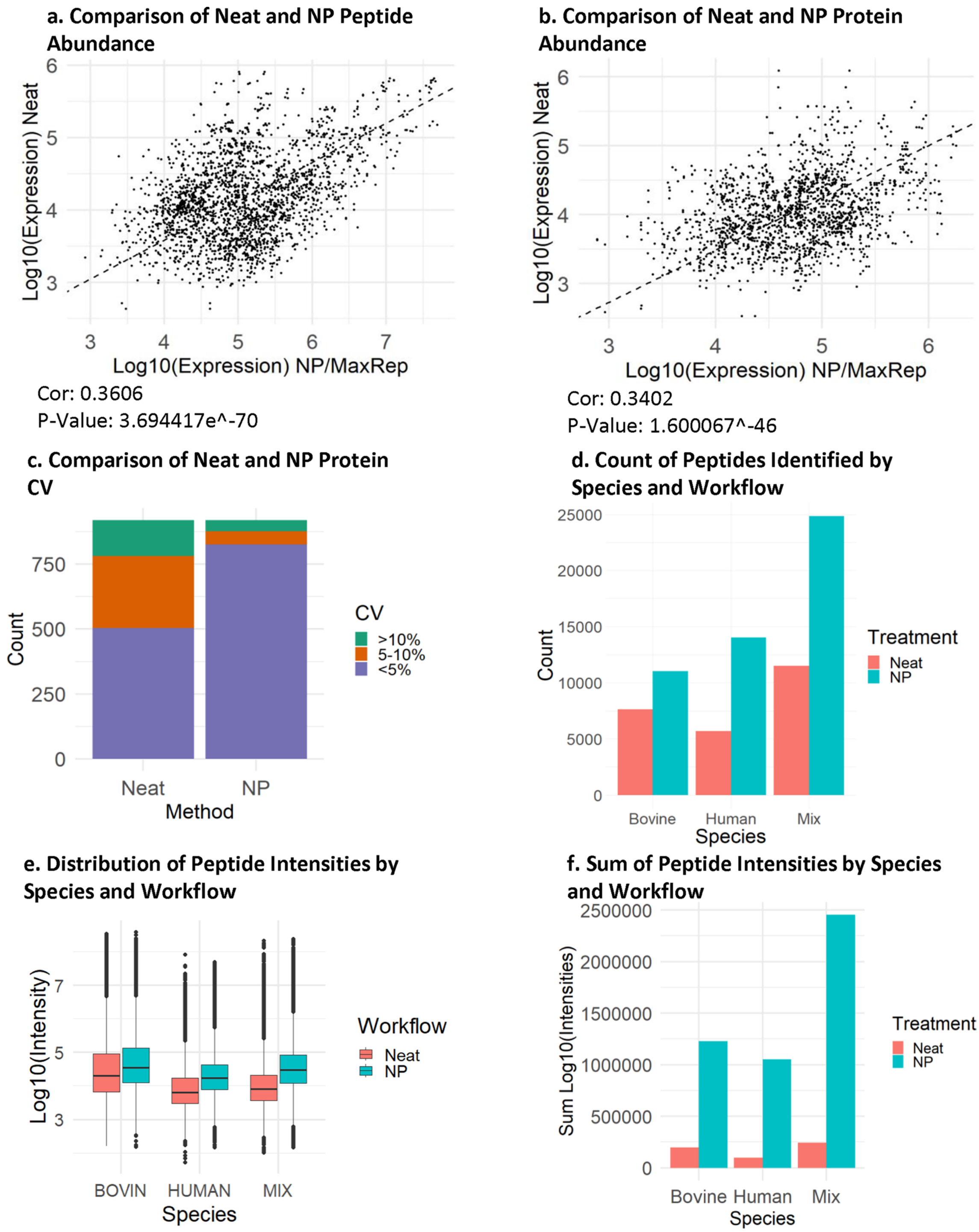

Extended Data Fig. 3 | NanoParticle Analysis. a, Peptide expression level is compared between quantification methods of 3 proliferating and 3 senescent monocyte samples, using NanoParticles or Neat (control). Only human peptides present in at least 4 of 6 samples, showing differential expression (Pval < 0.05, t-tests) in neat samples were included (n = 428). **b**, Protein expression level is compared between quantification methods of 3 proliferating and 3 senescent monocyte samples, using NanoParticles or Neat (control). Only human proteins present in at least 4 of 6 samples, and showing differential expression (Pval < 0.05, t-tests) in neat samples were included (n = 132) For all figures, human mapped peptides and proteins were used. **c**, CV (SD / Mean log10 intensities) was calculated for all proteins present in at least 2 of 3 Senescent samples and 2 of 3 Proliferating samples in Both Neat and NP MaxRep rollup (N = 919). **d**, The number of identified peptides in any sample in either NP or Neat treatments, classified as either human, bovine, or mixed. **e**, The distribution of the peptide log10 intensities sorted by species, quantified using either the standard (Neat) or Proteograph (NP) workflows. **f**, The sum of the log10 intensities sorted by species, quantified using either the standard (Neat) or Proteograph (NP) workflows.

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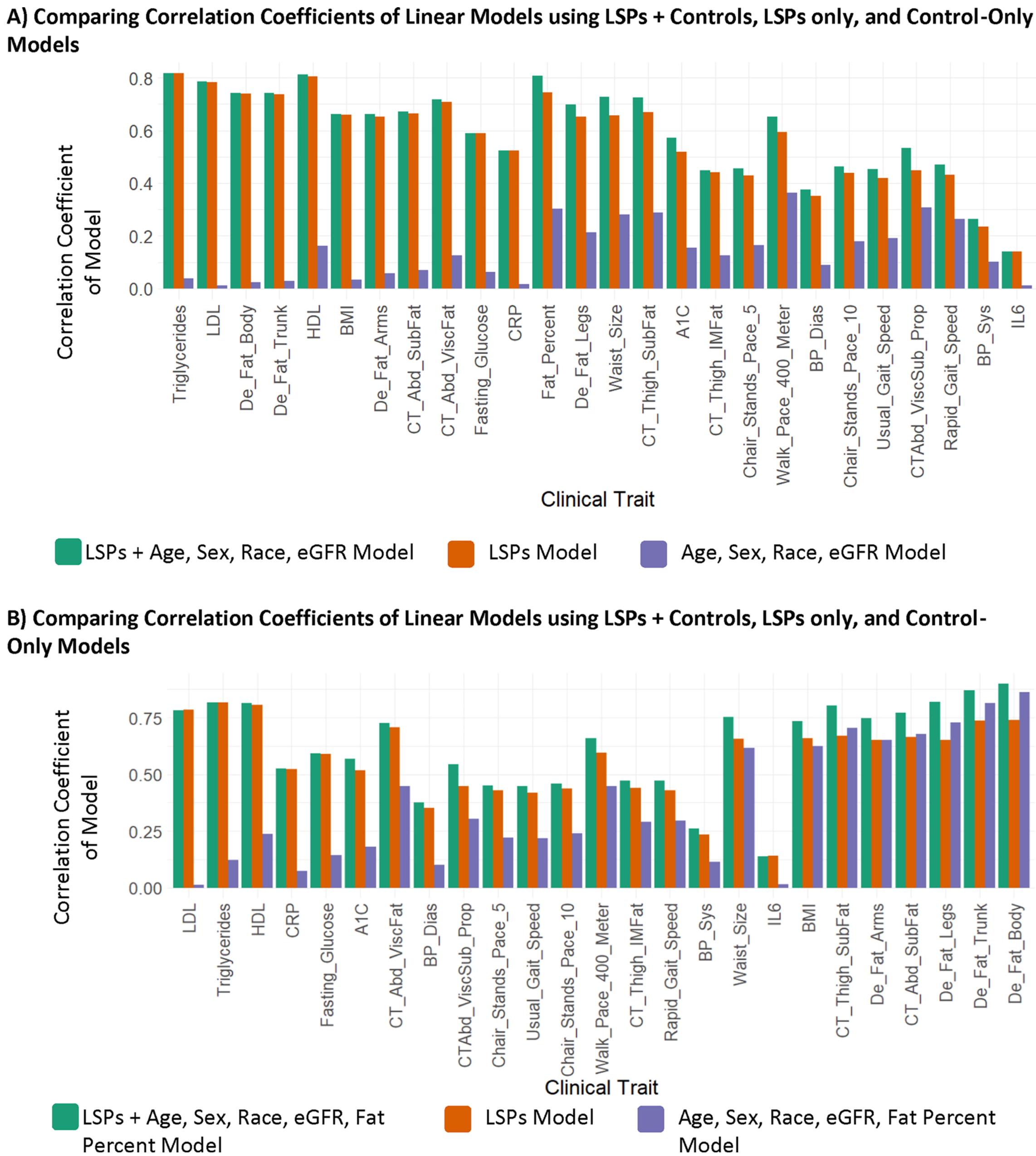

Extended Data Fig. 4 | LSPs Show Age-Independent Predictive Potential. **a**, Pearson linear models were constructed using covariates only (age, sex, race, and eGFR), LSPs only, or LSPs and covariates. Correlation coefficients for all models are shown by trait. **b**, This analysis is repeated but including fat percent as a covariate. Pearson linear models were constructed using covariates only (age, sex, race, and eGFR, and fat percent), LSPs only, or LSPs and covariates. Correlation coefficients for all models are shown by trait.

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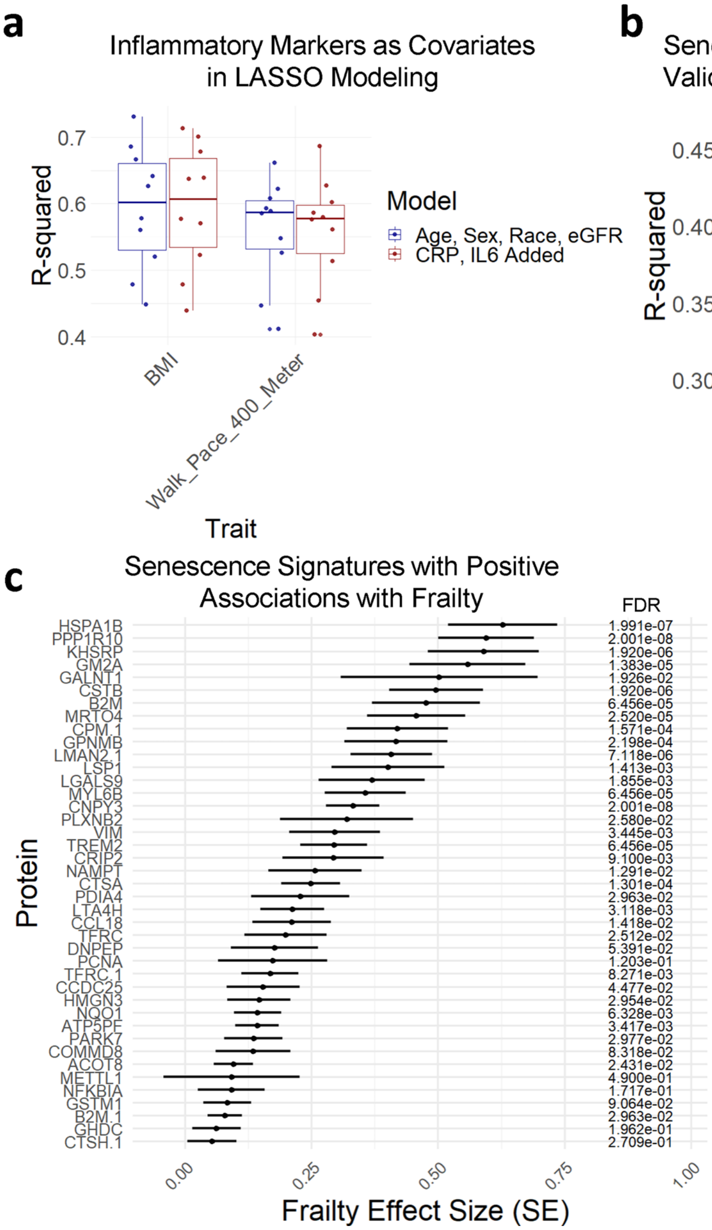

Extended Data Fig. 5 | Senescence Signatures Predict Frailty. a, LASSO modeling was used with age, sex, race, and eGFR as covariates in one model, and with CRP and IL-6 added as two additional covariates in a second model, for feature selection of proteins implicated in either BMI or walking pace. Selected features were used for trait prediction (90% train, 10% test). **b**, Senescence signatures were selected for a 44-component frailty index. Ten rounds of cross-validated (90% train, 10% test) trait prediction using a linear model with LSPs only was performed. Total sample sizes per trait are indicated in Table 1. **c**, Proteins chosen via machine learning and positively associated with frailty are shown by their association with the frailty index using linear modeling, with covariates age, sex, race, and eGFR.

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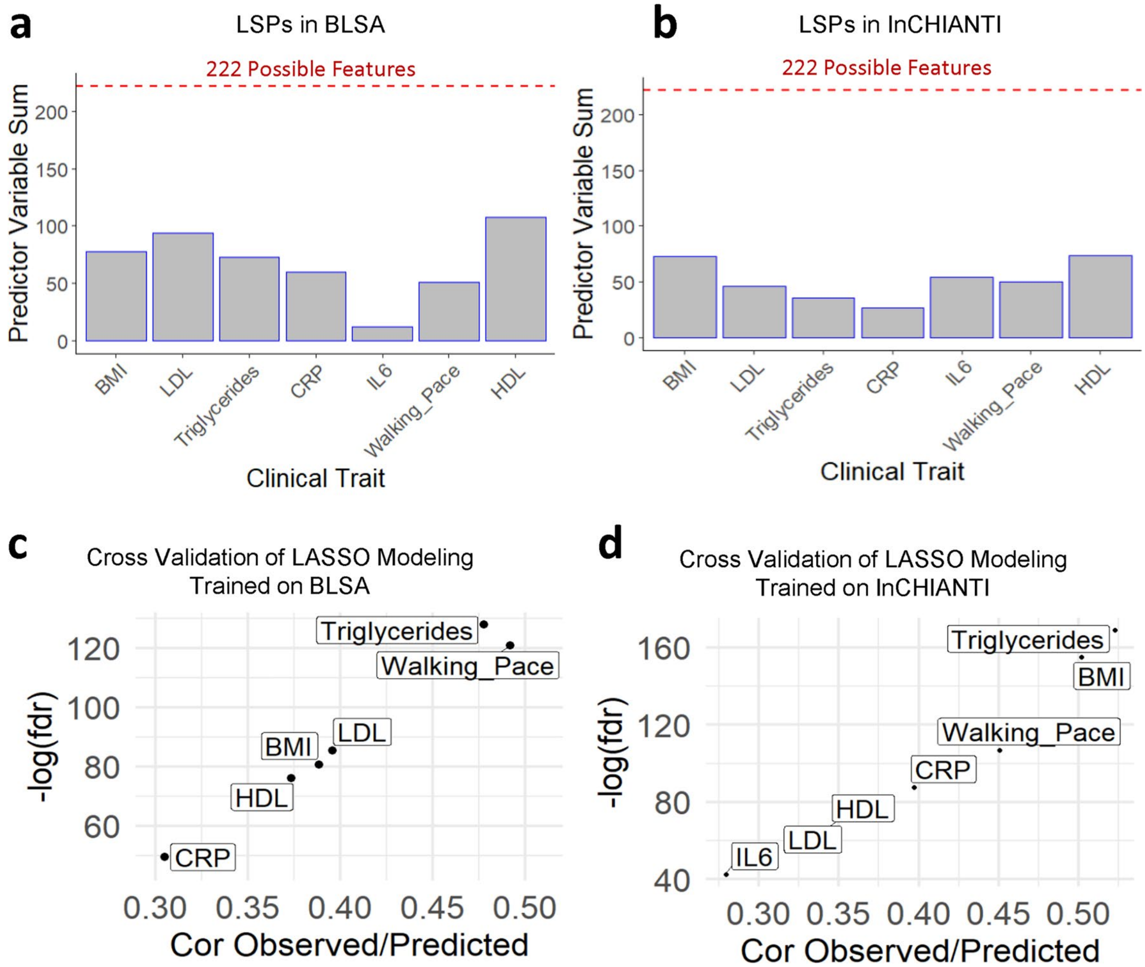

Extended Data Fig. 6 | LASSO-Selected Proteins by Cohort. a, 220 monocyte SASP were detected in both the BLSA (7k SomaScan) and Inchianti (1.3k SomaScan). LASSO modeling was used for feature selection in both Inchianti and BLSA. The number of LSPs selected via LASSO for each train in the BLSA are shown, including 220 SASP and 2 covariates (age and sex). **b**, LSPs selected in InCHIANTI, including 220 SASP and 2 covariates (age and sex). **c**, Linear models were constructed using only proteins selected in both studies for each trait. Spearman's correlation of predicted values of linear models trained on the BLSA and observed values in InCHIANTI. **d**, Spearman's correlation of predicted values of linear models trained on InCHIANTI and observed values in the BLSA.

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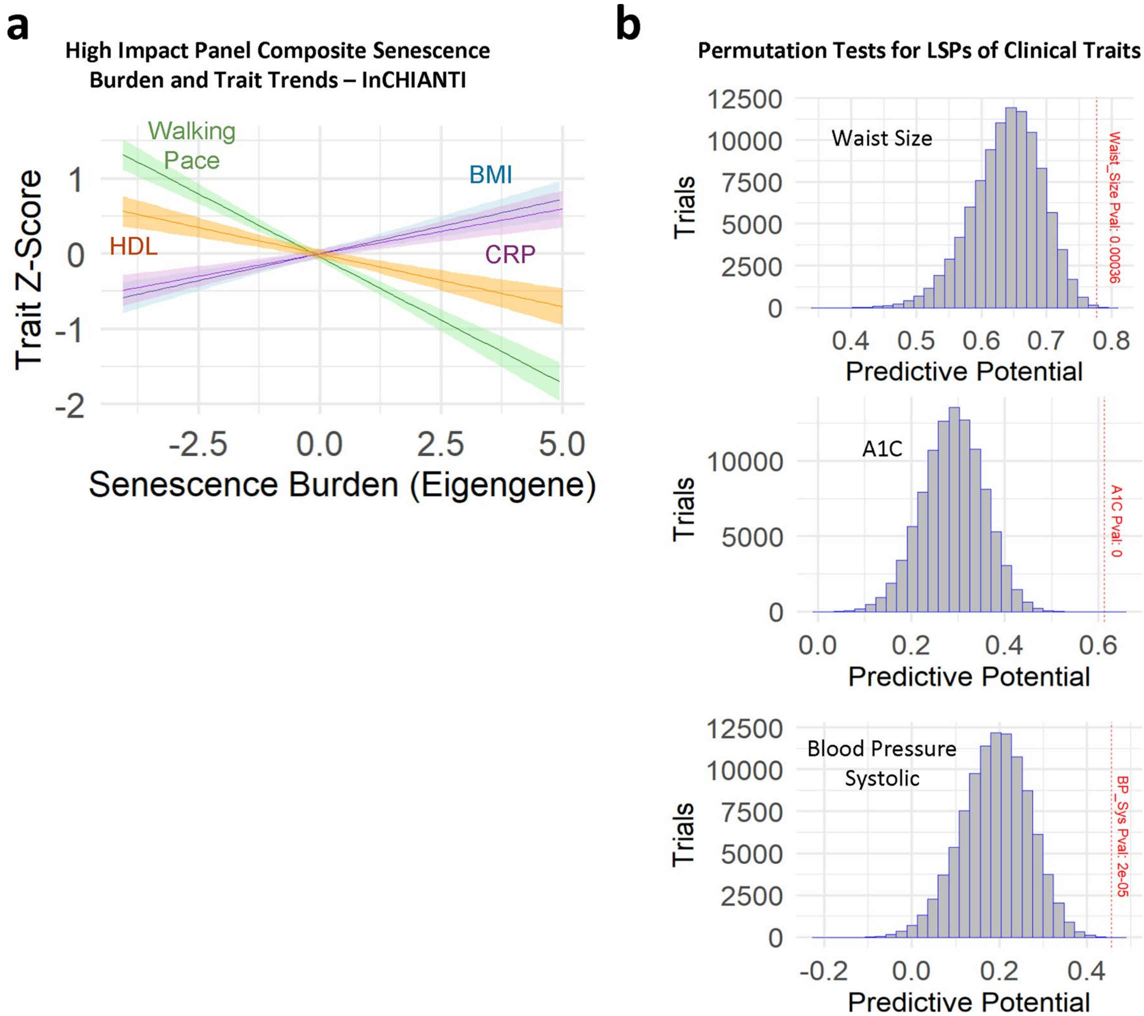

Extended Data Fig. 7 | Permutation Tests and LASSO Optimized Lambda Values. a, Principal-Component Analysis was used to condense the high-impact panel into a composite senescence burden score in the BLSA. Principal Component 1 was used to represent an eigengene for the high-impact panel. With the InCHIANTI cohort ranked from low to moderate to high senescence burden, linear trait trends reveal that positive traits HDL and Walking Pace show a negative trend, while negative traits BMI and CRP show a positive trend. **b**, Permutation tests comparing the predictive potential is shown for LSPs compared with randomly selected proteins. Linear models for each trait were created either using LSPs or randomly selected proteins of the same size. Models were trained on 80% of the data and used to predict the clinical traits for the remaining 20%. Randomly selected proteins models were trained and tested 100,000 times per trait and compared with the accuracy of the LSP-only model. Red dotted lines show where the Spearman's correlation of the LSP-only model lies in relation to the bell curve for the randomly selected protein models.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Available in the supplement.
Data analysis	GraphPad Prism (version v10.3.0) R (version 4.2.0) Rstudio (version 2023.06.0-421) R scripts: - https://github.com/geroproteomics/EN_Repeat - https://github.com/geroproteomics/EN_Test R Packages - glmnet (version 4.1-6) - ClusterProfiler (version 4.6.0)

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All raw mass spectrometry data files and associated quantitative and statistical reports, metadata, and supplemental data are available on MassIVE (dataset identifier: MSV000095315). FTP download link: ftp://massive.ucsd.edu/v08/MSV000095315/. All the aggregated phenotype data has been provided as source data as well as supplementary tables. To comply with patient consent and data sharing agreements, researchers are welcome and encouraged to request use of more detailed BLSA data for scientific projects by developing a pre-analysis plan that can be submitted for approval (https://blsa.nia.nih.gov/how-apply).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Reported throughout manuscript and Table 1 and Table S3
Reporting on race, ethnicity, or other socially relevant groupings	Reported throughout manuscript and Table 1 and Table S3
Population characteristics	Reported throughout manuscript and summarized demographics details in Table 1 and Table S3
Recruitment	BLSA and InCHIANTI studies recruitment protocols are cited in the methods section.
Ethics oversight	All human studies were compliant with ethical regulations. The BLSA protocol (03AG0325) was approved by the institutional review board of the National Institute of Environmental Health Science, part of the National Institutes of Health. The InCHIANTI study (exemption #11976) protocol was approved by Medstar Research Institute (Baltimore, Maryland), the Italian National Institute of Research and Care of Aging Institutional Review, and the Internal Review Board of the National Institute for Environmental Health Sciences (NIEHS). All participants in both studies provided written informed consent and were not compensated.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Cohorts were previously selected from the BLSA and InCHIANTI for plasma proteomics analysis using the SomaScan Assay, and are sufficiently large (N = 1060 and 997, respectively) to identify meaningful and statistically significant clinical insights based on determinations made in our previously published study (Tanaka et al. 2020. eLife).
Data exclusions	To avoid possible confounding effects of medication use, patients taking diabetes or hypertension medication had their A1C or blood pressure values excluded from statistical analysis.
Replication	Replication of findings are reported between BLSA and InCHIANTI, two independent aging studies. For several clinical parameters (IL6, LDL, HDL, CRP, Walking Pace, BMI, and Triglycerides), models trained on one cohort and used to predict the trait on the second cohort showed significant replication ability.
Randomization	- There is no allocation of samples into treatment groups in this study. - Randomization was used in permutation tests (to evaluate predictive power of random proteins vs SASP) and for mass spectrometry samples run order. - During feature selection (lasso modeling), covariates sex, age, race, eGFR (kidney function) were supplied for BLSA and age and sex were supplied with InCHIANTI, with the purpose of selecting features that show significant association with traits of interest irrespective of the covariates. - During validation, models were trained on randomly selected 80% of each cohort and used to predict clinical traits on the remaining 20%.

Blinding	BLSA and InCHIANTI are observational studies, therefore no blinding is used.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Supplier, catalog, clone, lot anti-human CD14-PE (BioLegend, San Diego, CA; #325606, Clone: HCD14, Lot# B188763) anti-human CD11b-BV605 (BioLegend, San Diego, CA; #301331, Clone: ICRF44, CD11b Lot# B184305)
Validation	The vendor for anti-human CD14-PE (BioLegend, San Diego, CA; #325606) reports verified reactivity to Human (https://www.biolegend.com/en-gb/products/pe-anti-human-cd14-antibody-3952) The vendor for anti-human CD11b-BV605 reports verified reactivity to Human (BioLegend, San Diego, CA; #301331,(https://www.biolegend.com/nl-nl/products/brilliant-violet-605-anti-human-cd11b-antibody-8493)

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	THP-1 monocyte cell line (ATCC), derived from 1 year old male
Authentication	Cells were authenticated by flow cytometry for expression of classical monocyte markers
Mycoplasma contamination	Cells were free of Mycoplasma, as determined by the MycoAlert Kit (Lonza #LT07-218)
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines are used in the study.

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	NCT00233272
Study protocol	BLSA protocol - 03AG0325 (approved by NIH IRB); InCHIANTI study - exemption #11976 (protocol approved by NIH IRB & Medstar Research Institute
Data collection	The BLSA study is a population-based study that started in 1958 to evaluate the contributors of healthy aging in subjects recruited from the DC/Baltimore metropolitan area that are 20 years old and older 53-55. It involves a data collection of clinical parameters, such as waist circumference, BMI and blood pressure that are assessed during a standard medical exam. Blood tests were performed at a Clinical Laboratory Improvement Amendments certified clinical laboratory at Medstar Harbor Hospital, Baltimore, MD. Total cholesterol was measured using alkaline phosphatase, HDL and LDL with dextran magnetic beads, triglycerides with colorimetric methods, glucose with glucose oxidase using the Vitros system (Ortho Clinical Diagnostics, Raritan, NJ). Serum inflammatory markers IL6 (R&D System, Minneapolis, MN) and CRP (Alpco, Salem, NH) were measured with enzyme-linked immunosorbent assay (ELISA). HbA1C levels were measured using liquid chromatography by an automated DiaSTAT analyzer (Bio-Rad, Oakland, CA). Grip strength was measured three times on each of the right and left hand and the highest average grip strength was reported. Usual gait speed was measured in two trials of a 6-m walk and the faster time between the two trials is used for analysis. Body fat measurements were made using CT/dual x-ray absorptiometry scans. To avoid possible confounding effects of medication use, patients taking

diabetes or hypertension medication had their A1C or blood pressure values excluded from statistical analysis.

The InCHIANTI is a similar population-based study of aging conducted in the Chianti region of Tuscany, Italy previously described in more detail (Ferrucci et al, 2000, J Am Geriatr Soc). Residents from the population registry of Greve in Chianti (a rural area) and Bagno a Ripoli (near Florence) ranging in age from 21 to 102 years participated in the study.

Outcomes

These are observational studies. The method of collection of clinical phenotypes are described in the data collection section above.

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.